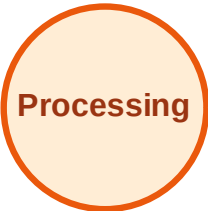


BFS Traversal

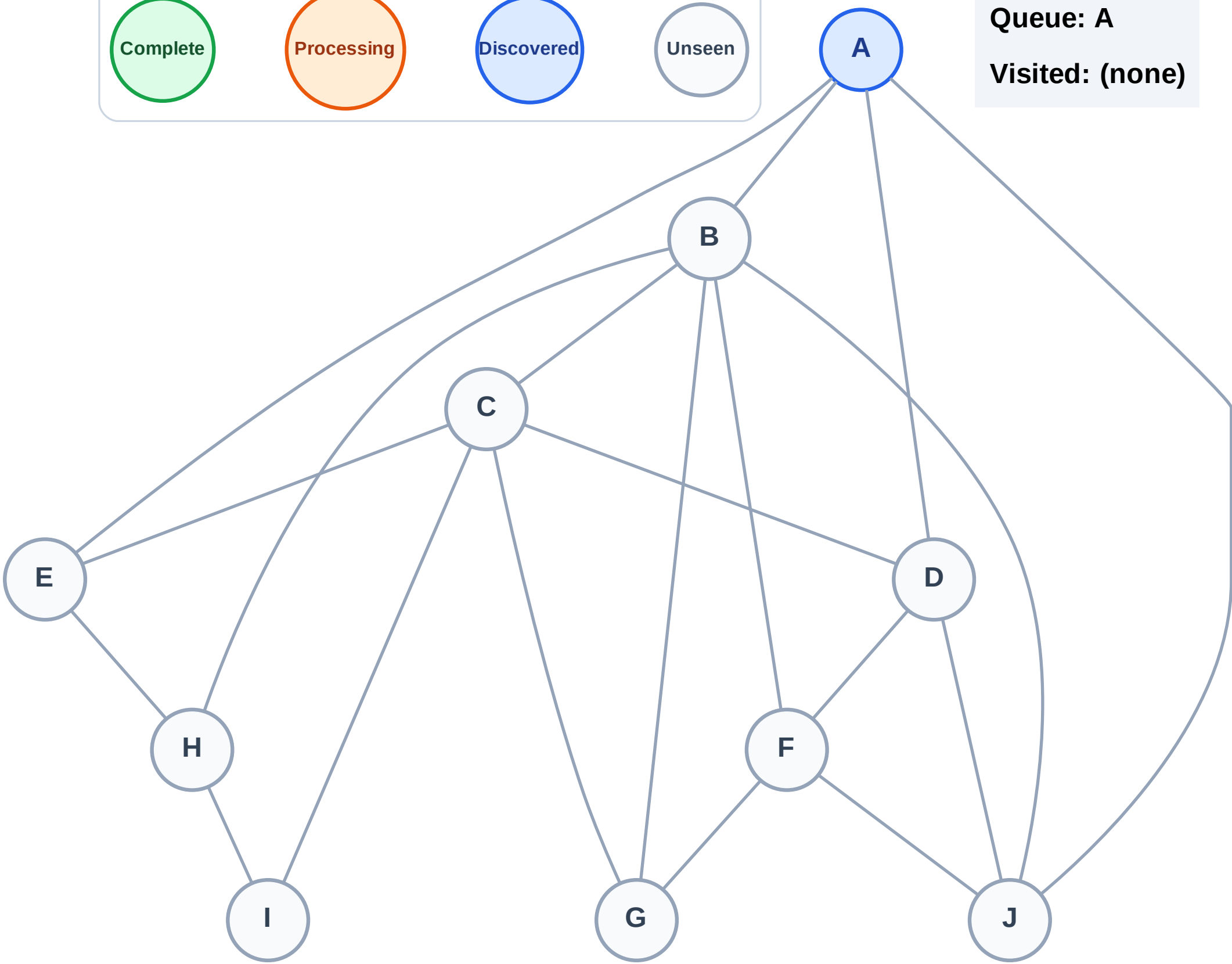
Step 1: Start at A

Legend



Queue: A

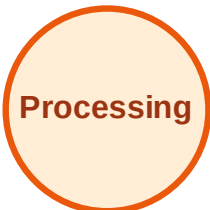
Visited: (none)



BFS Traversal

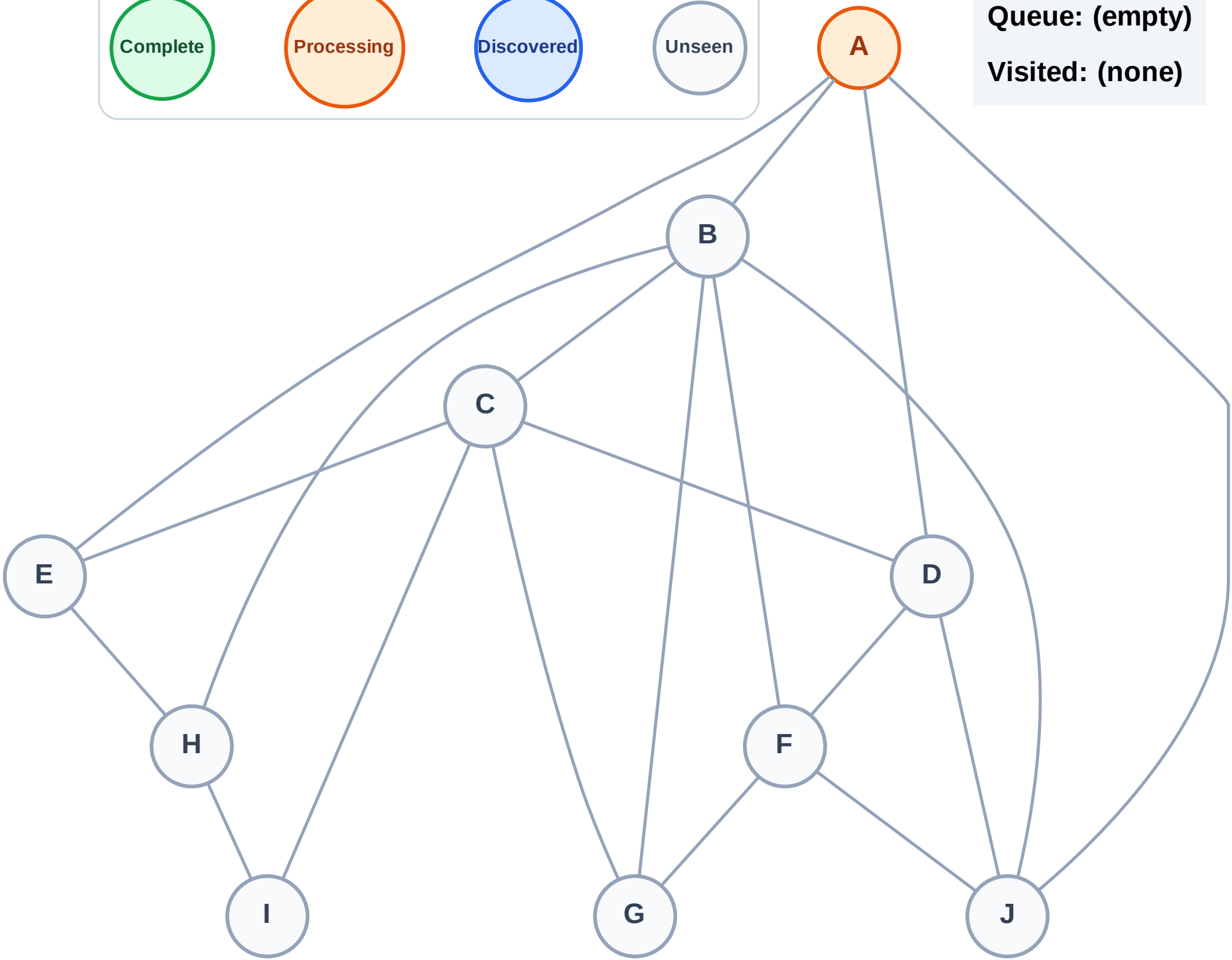
Step 2: Process node A

Legend



Queue: (empty)

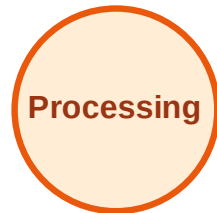
Visited: (none)



BFS Traversal

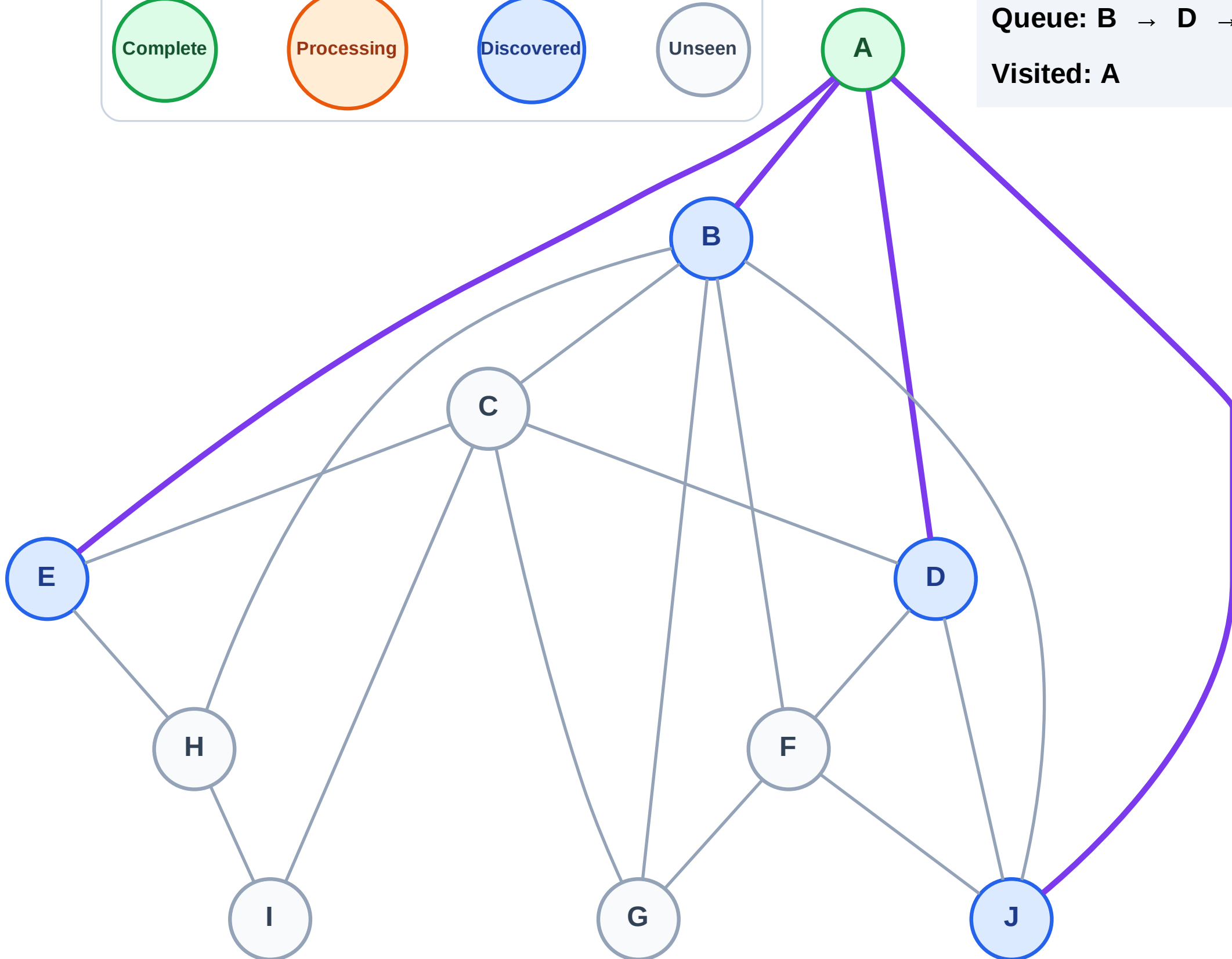
Step 3: Discover B, D, E, J from A

Legend



Queue: B → D → E → J

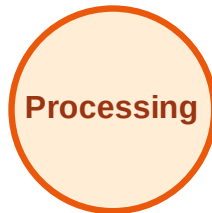
Visited: A



BFS Traversal

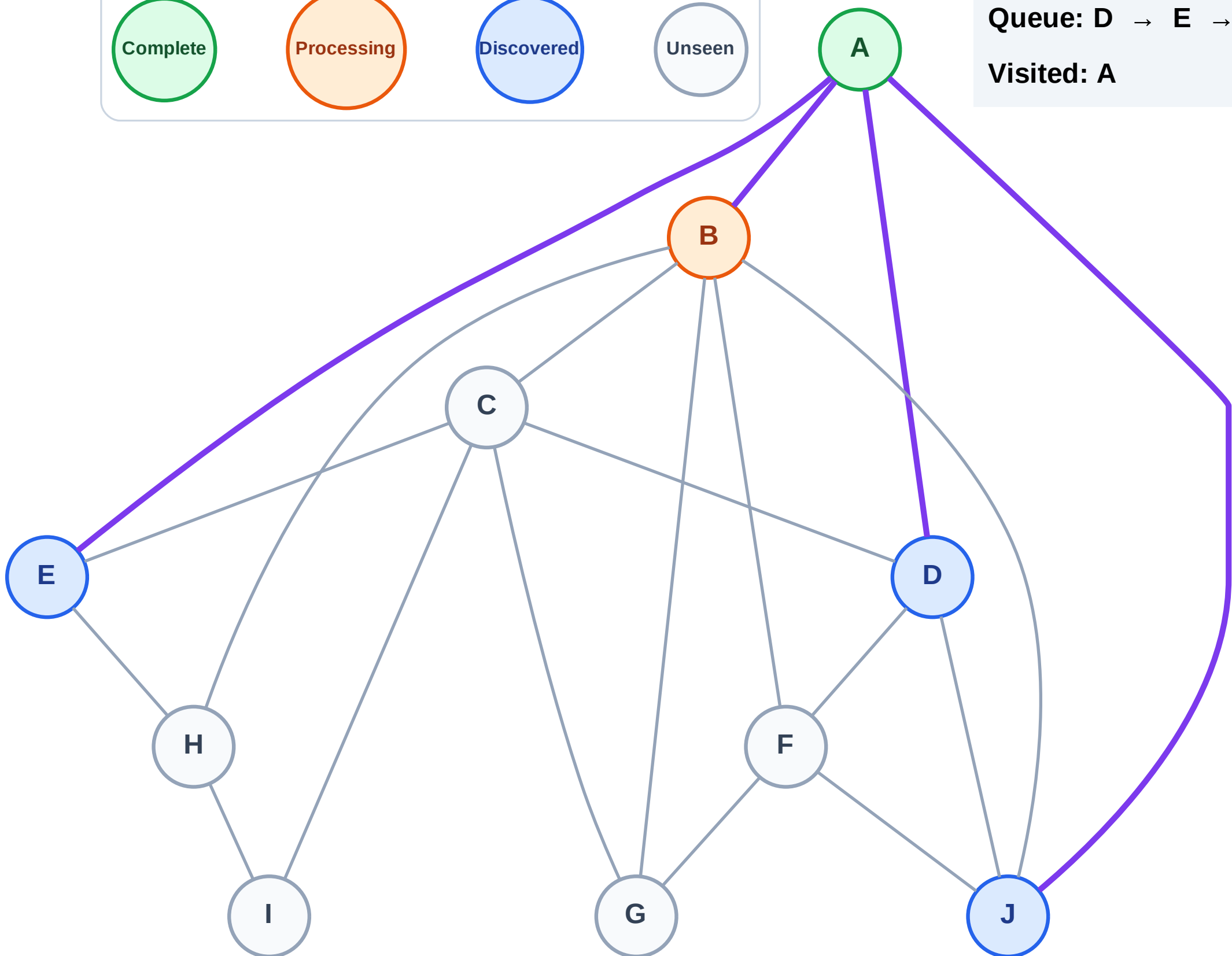
Step 4: Process node B

Legend



Queue: D → E → J

Visited: A



BFS Traversal

Step 5: Discover C, F, G, H from B

Legend

Complete

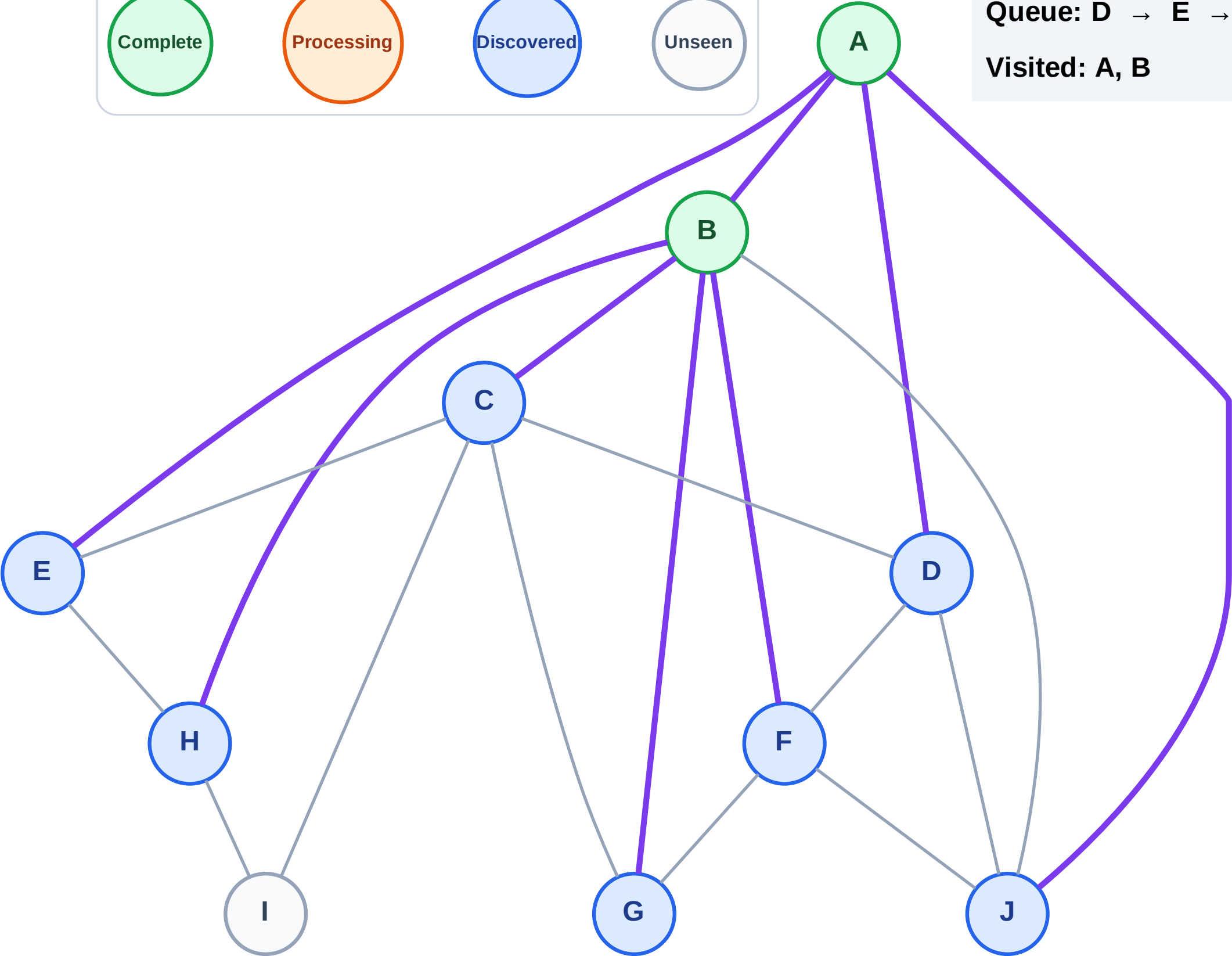
Processing

Discovered

Unseen

Queue: D → E → J → C → F → G → H

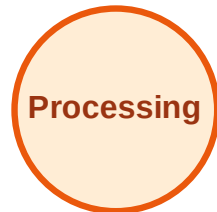
Visited: A, B



BFS Traversal

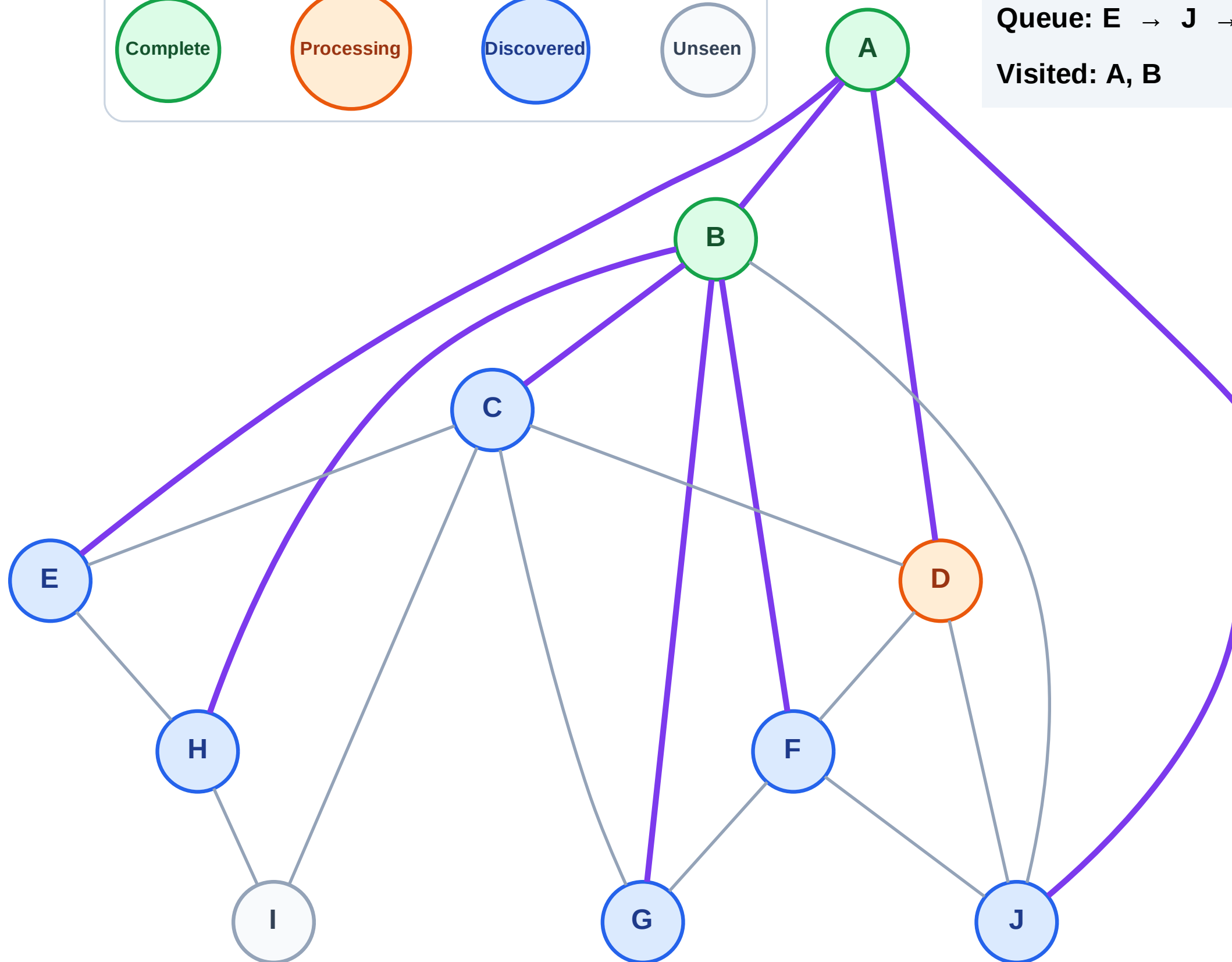
Step 6: Process node D

Legend



Queue: E → J → C → F → G → H

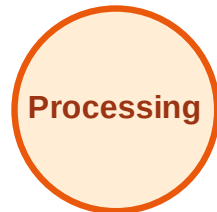
Visited: A, B



BFS Traversal

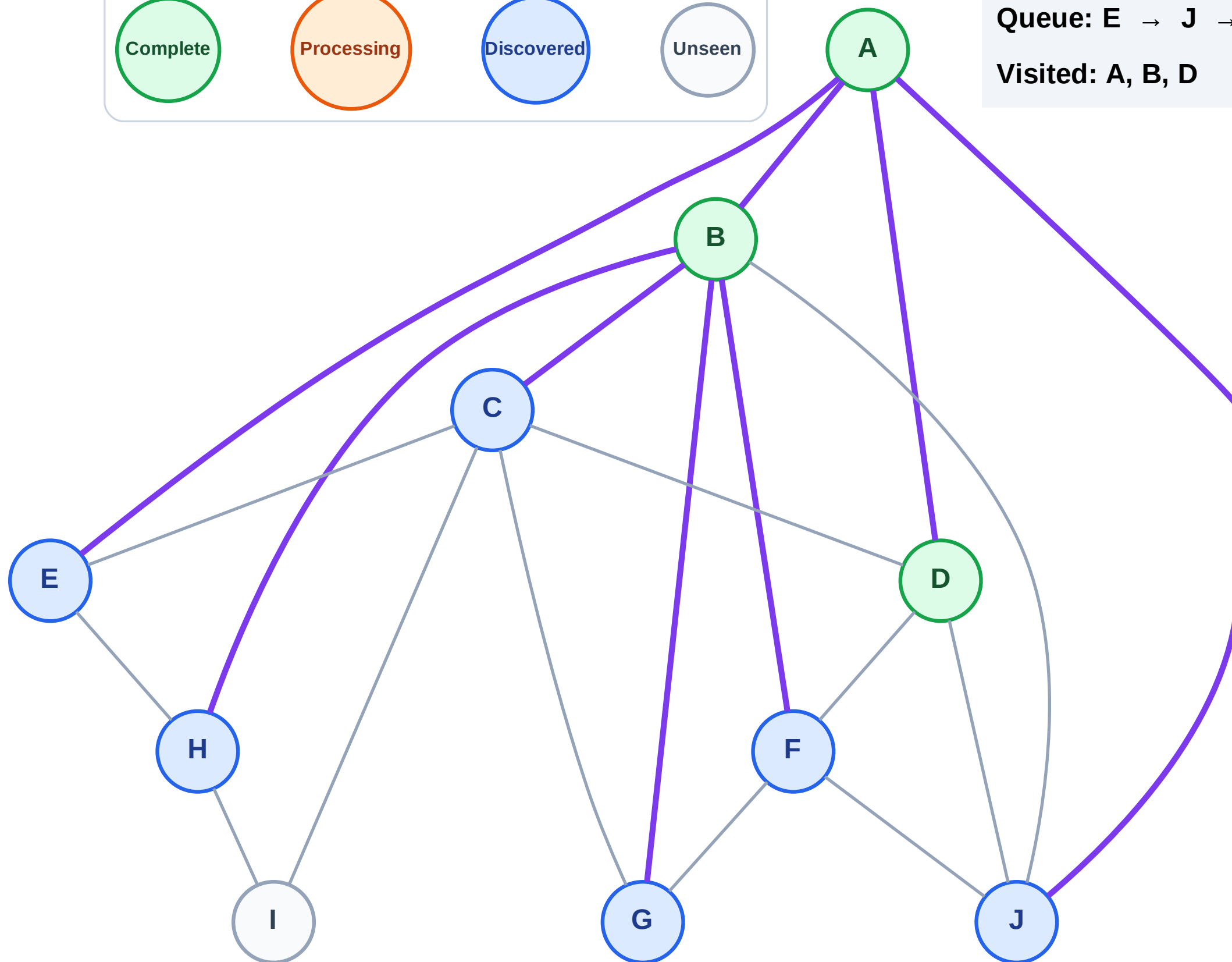
Step 7: No new neighbors from D

Legend



Queue: E → J → C → F → G → H

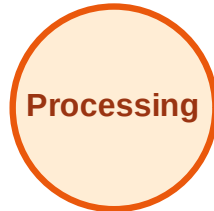
Visited: A, B, D



BFS Traversal

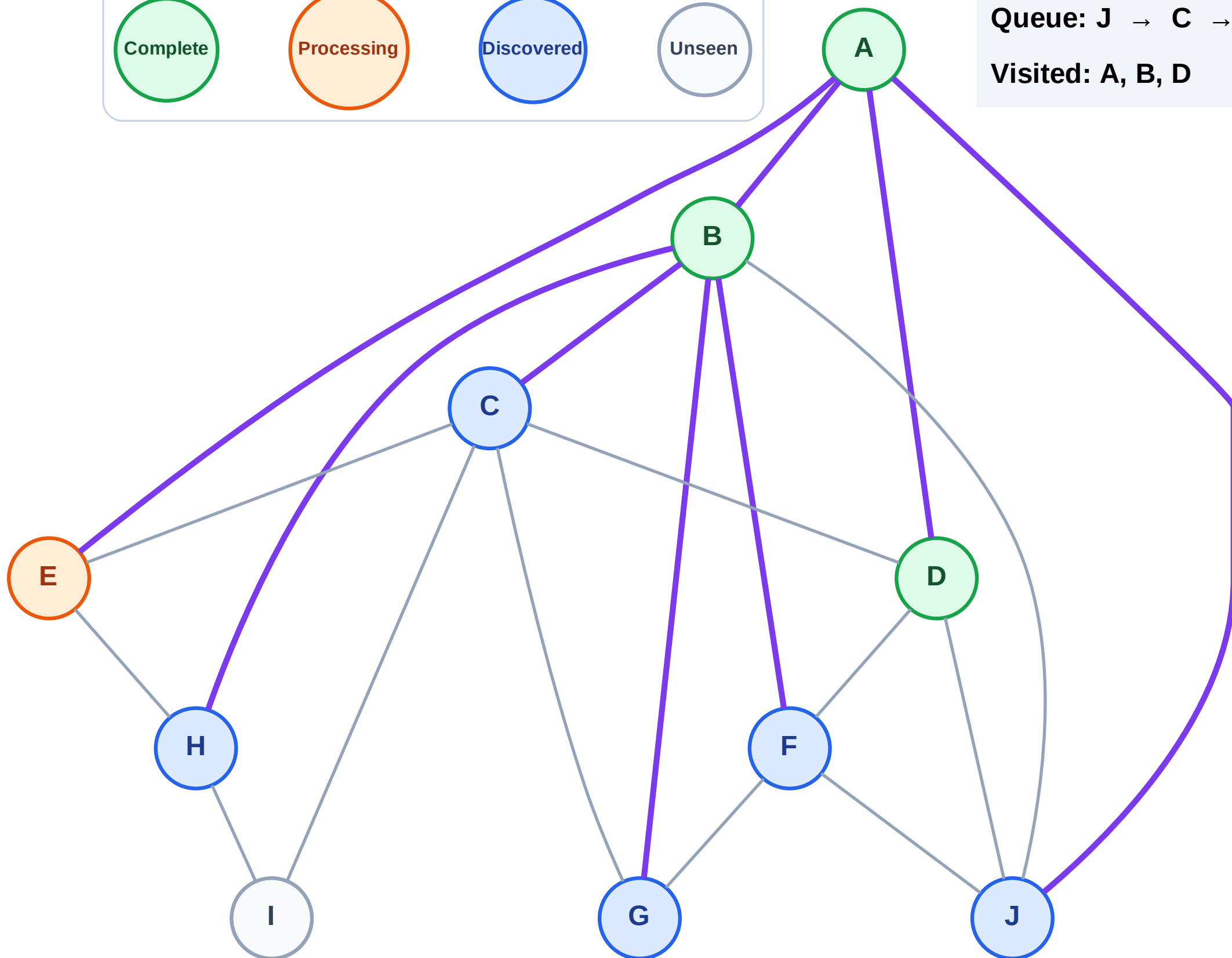
Step 8: Process node E

Legend



Queue: J → C → F → G → H

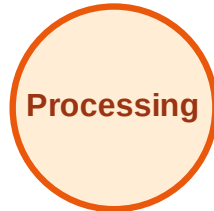
Visited: A, B, D



BFS Traversal

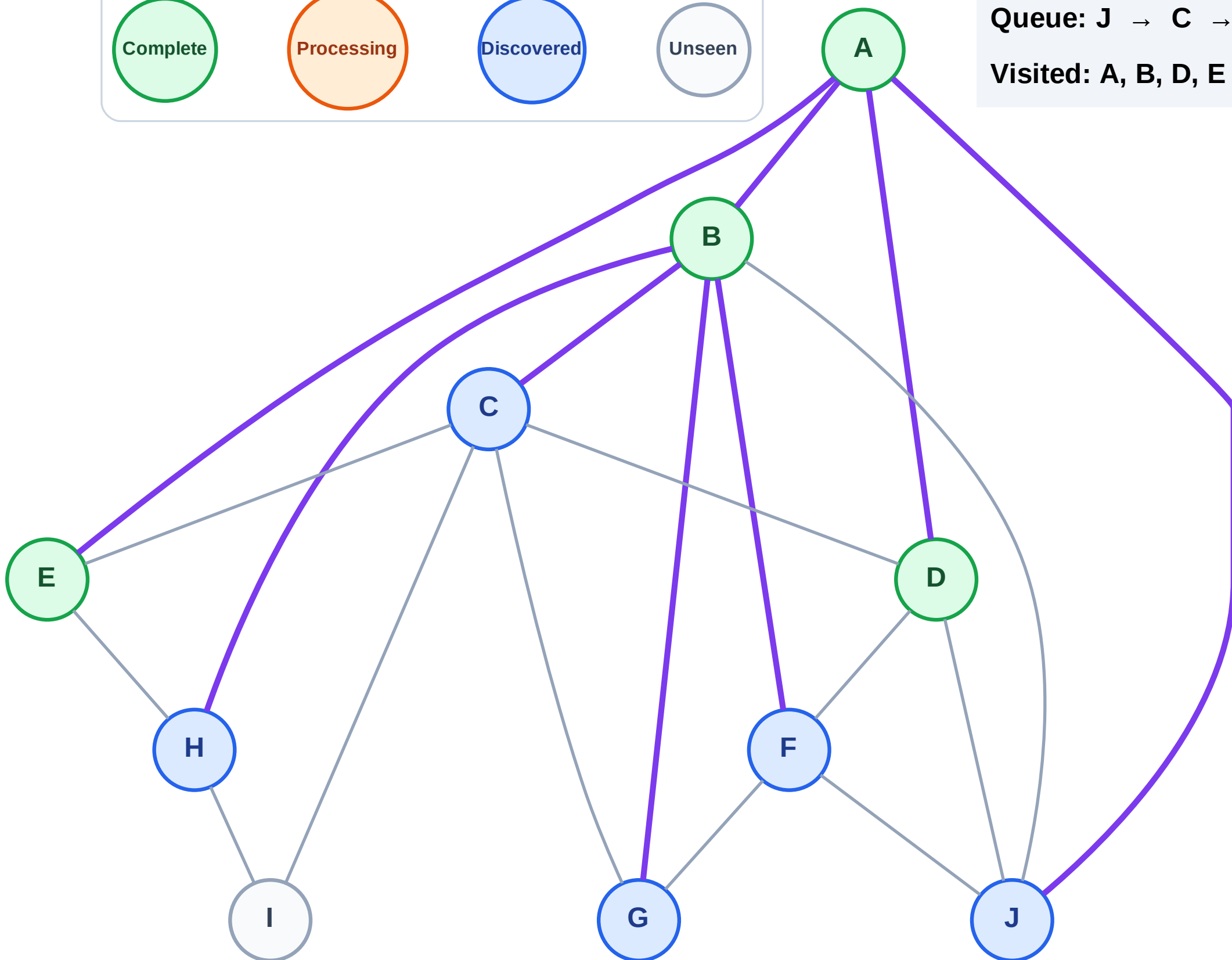
Step 9: No new neighbors from E

Legend



Queue: J → C → F → G → H

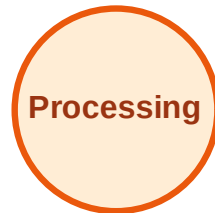
Visited: A, B, D, E



BFS Traversal

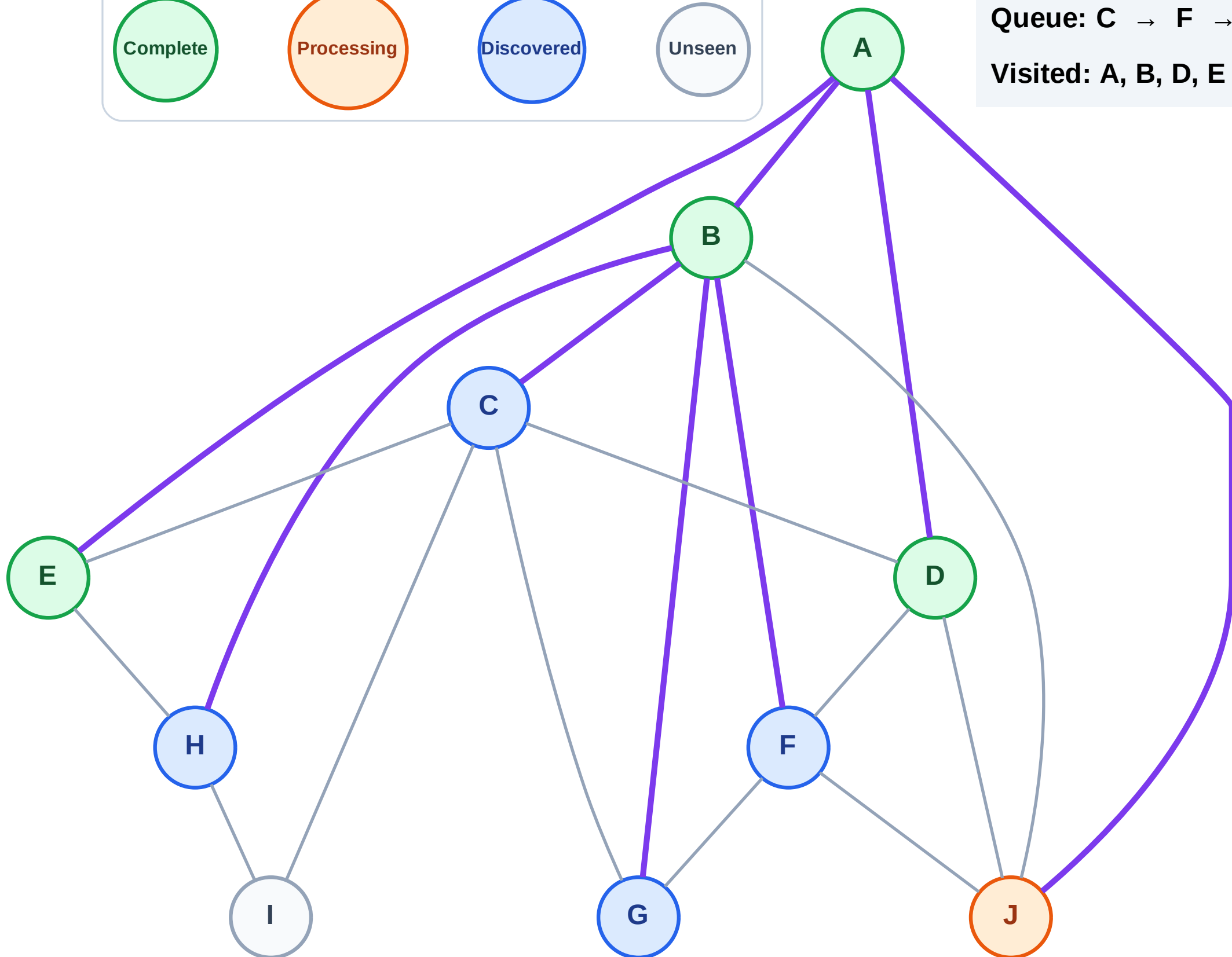
Step 10: Process node J

Legend



Queue: C → F → G → H

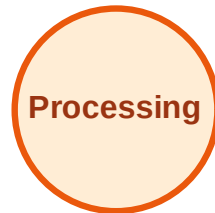
Visited: A, B, D, E



BFS Traversal

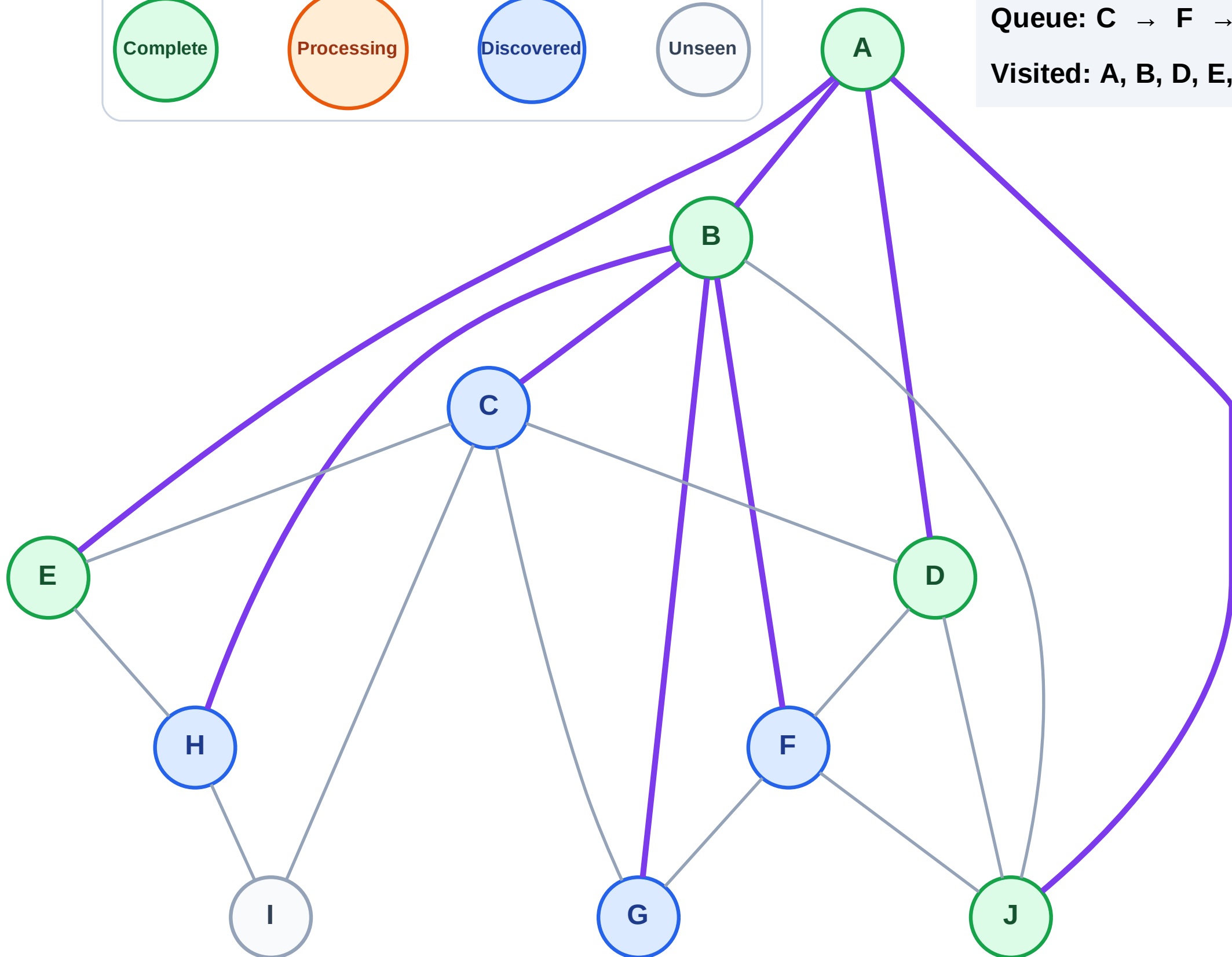
Step 11: No new neighbors from J

Legend



Queue: C → F → G → H

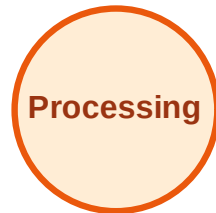
Visited: A, B, D, E, J



BFS Traversal

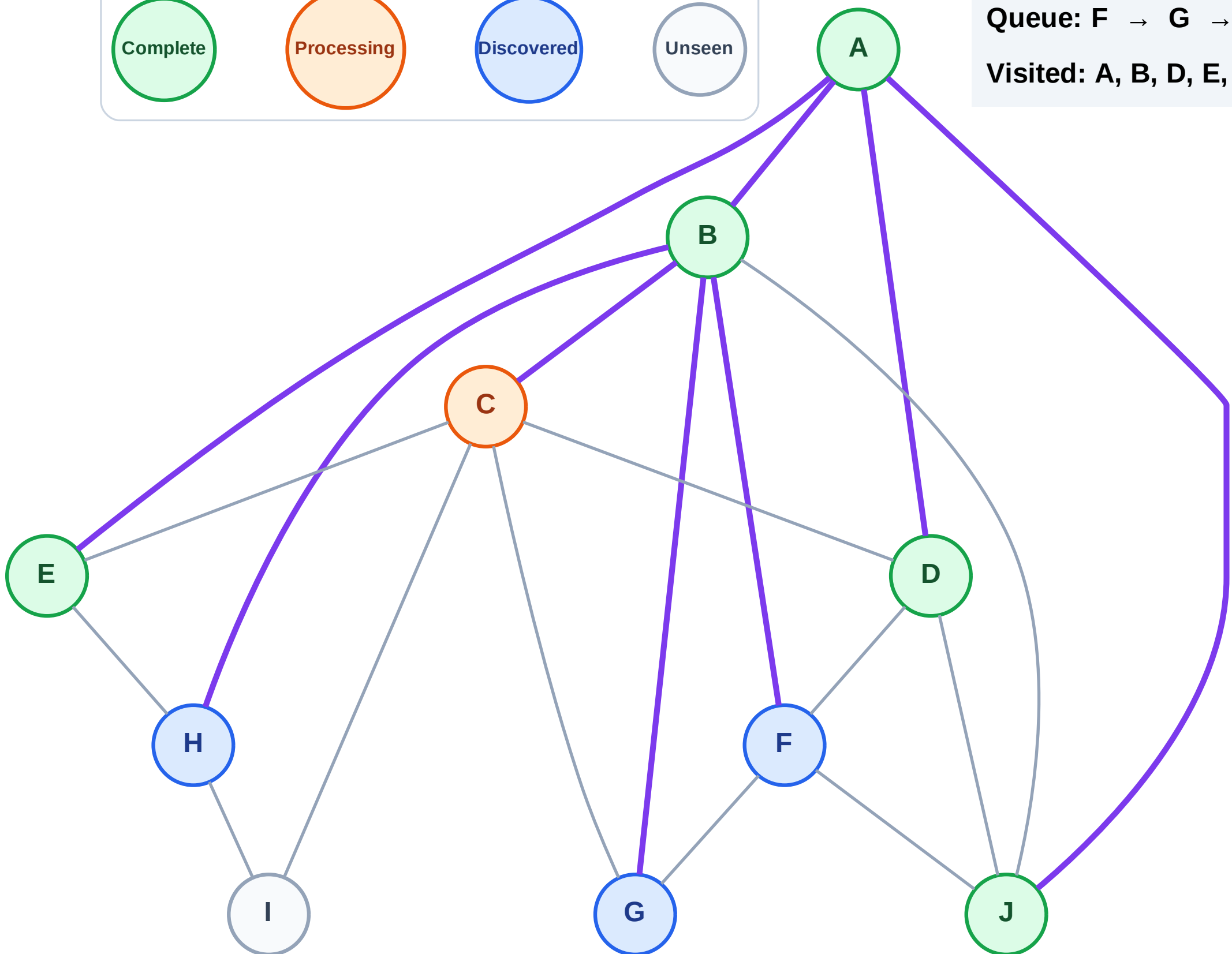
Step 12: Process node C

Legend



Queue: F → G → H

Visited: A, B, D, E, J

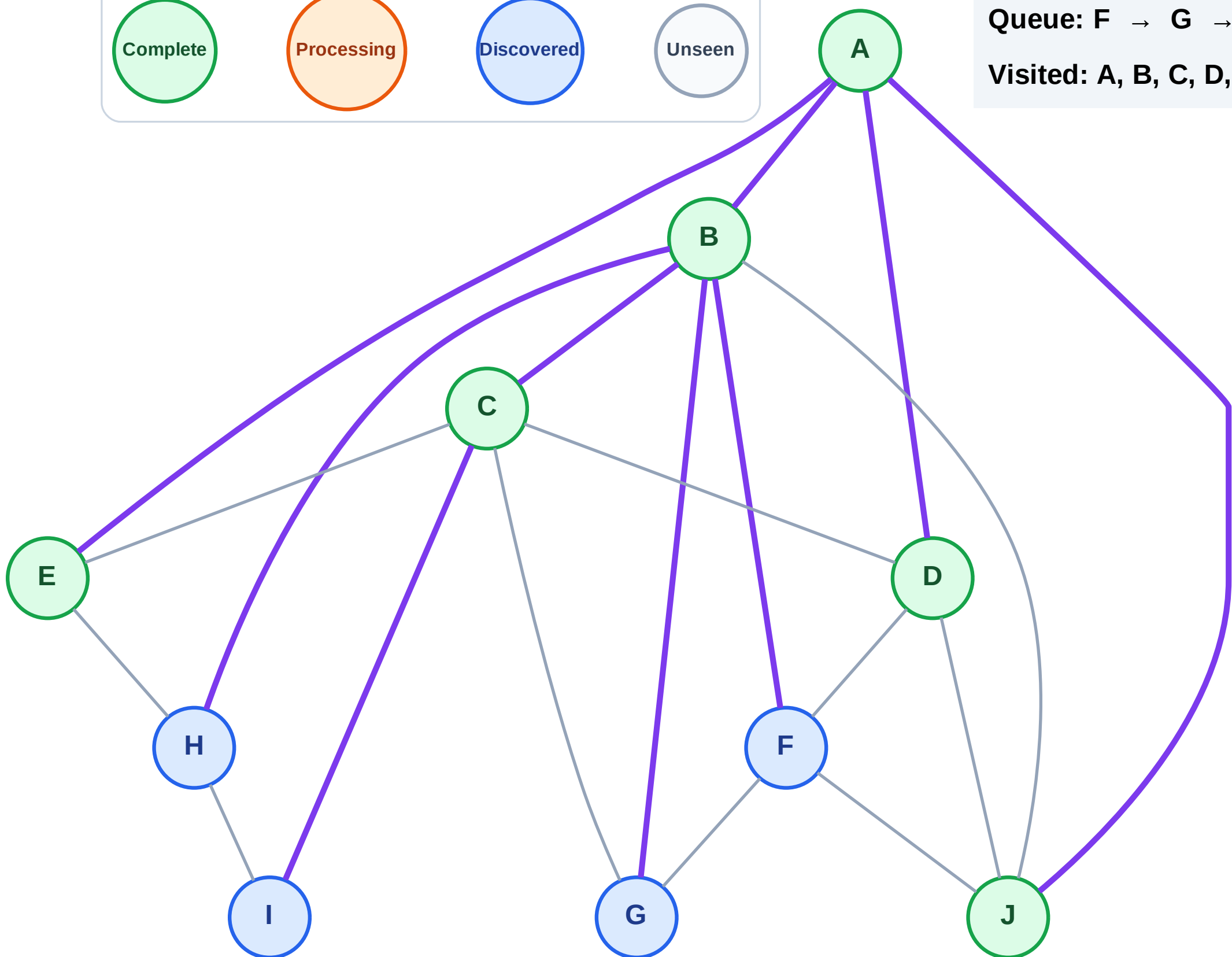


Step 13: Discover I from C

Legend

Unseen

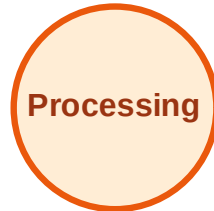
Visited: A, B, C, D, E, J



BFS Traversal

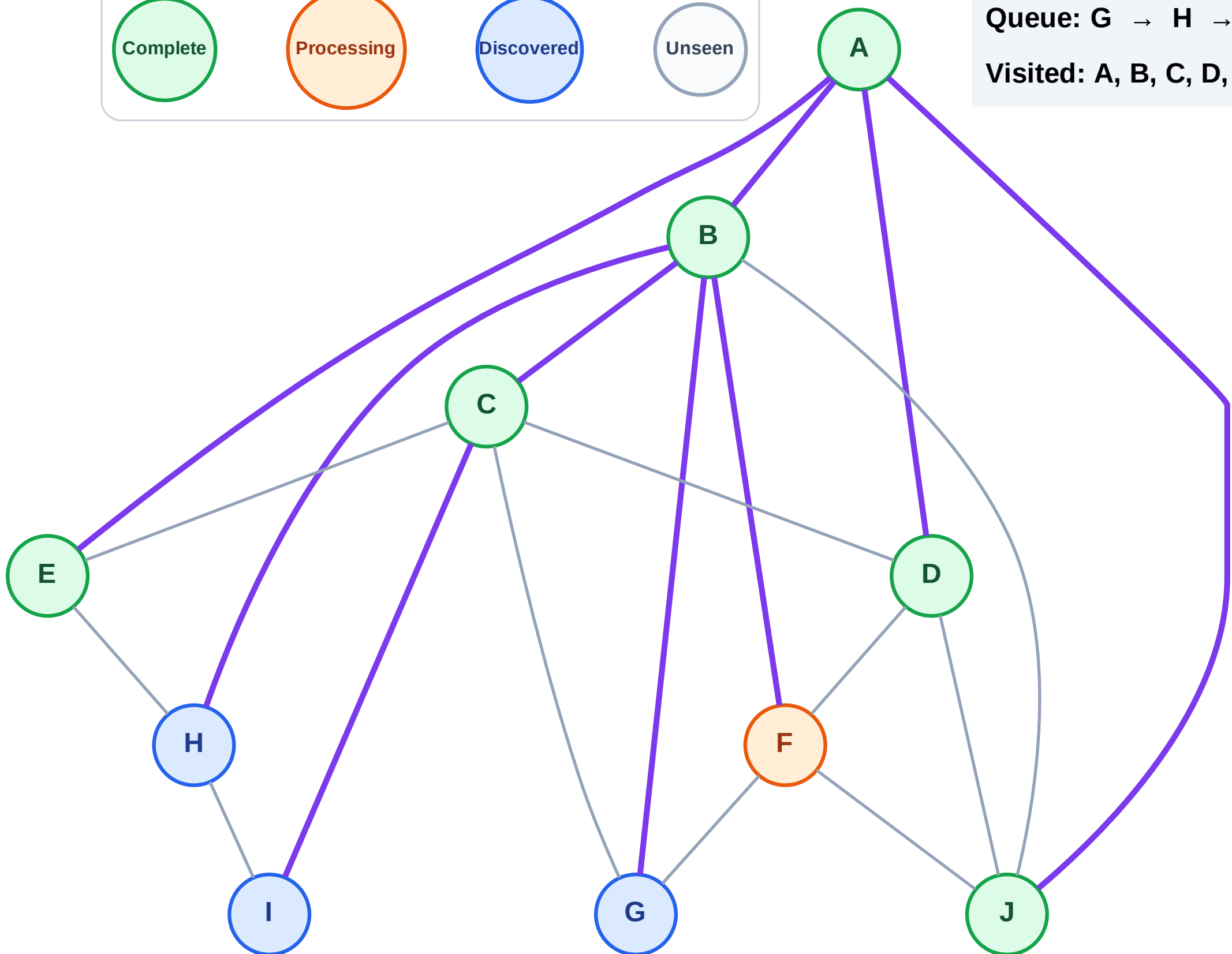
Step 14: Process node F

Legend



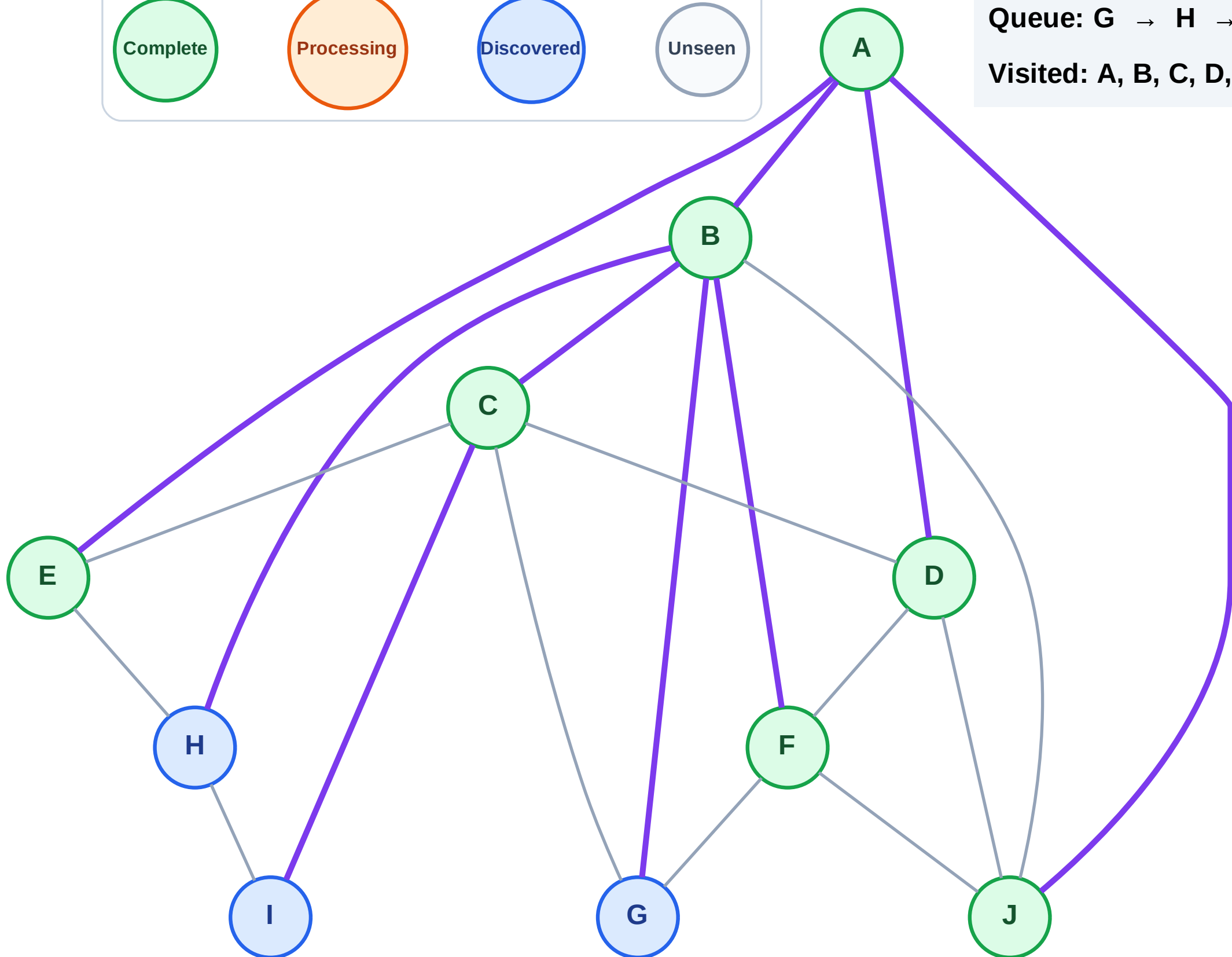
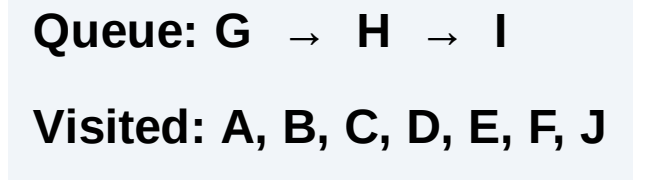
Queue: G → H → I

Visited: A, B, C, D, E, J



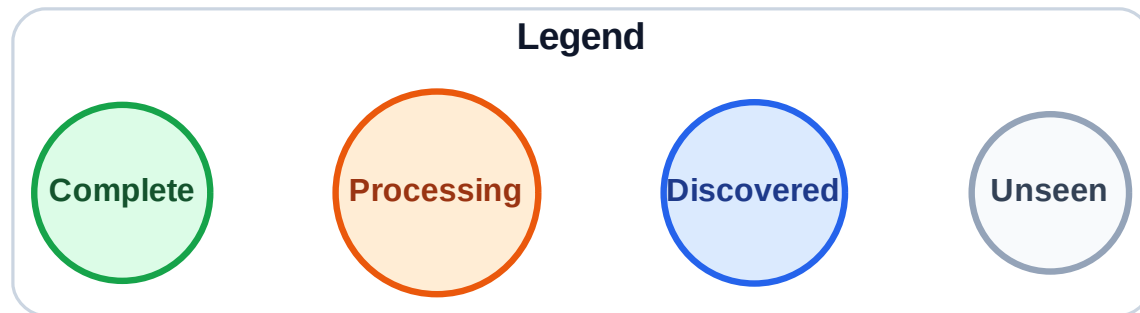
Step 15: No new neighbors from F

Legend



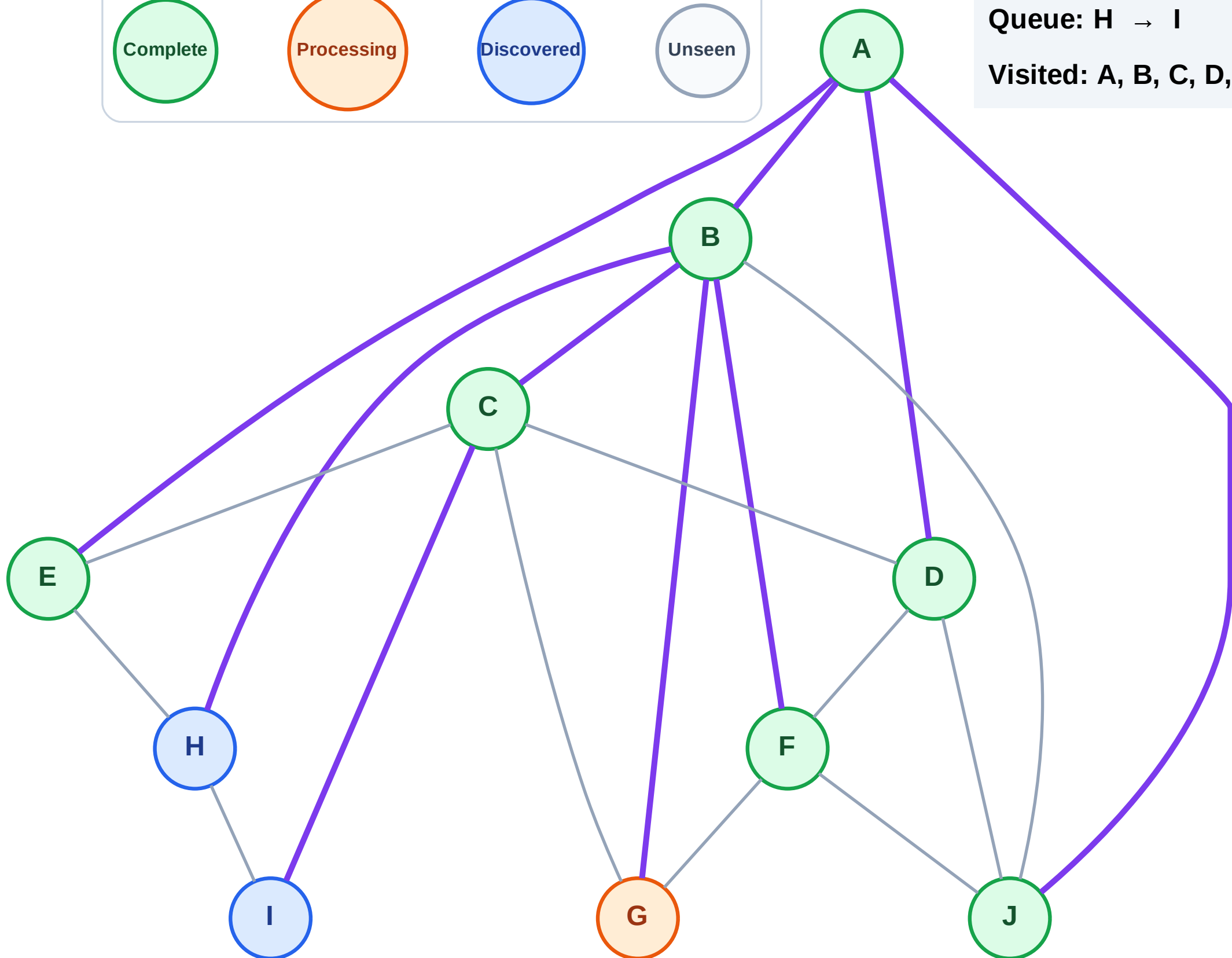
Step 16: Process node G

Step 16: Process node G



Queue: H → I

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 17: No new neighbors from G

Legend

Complete

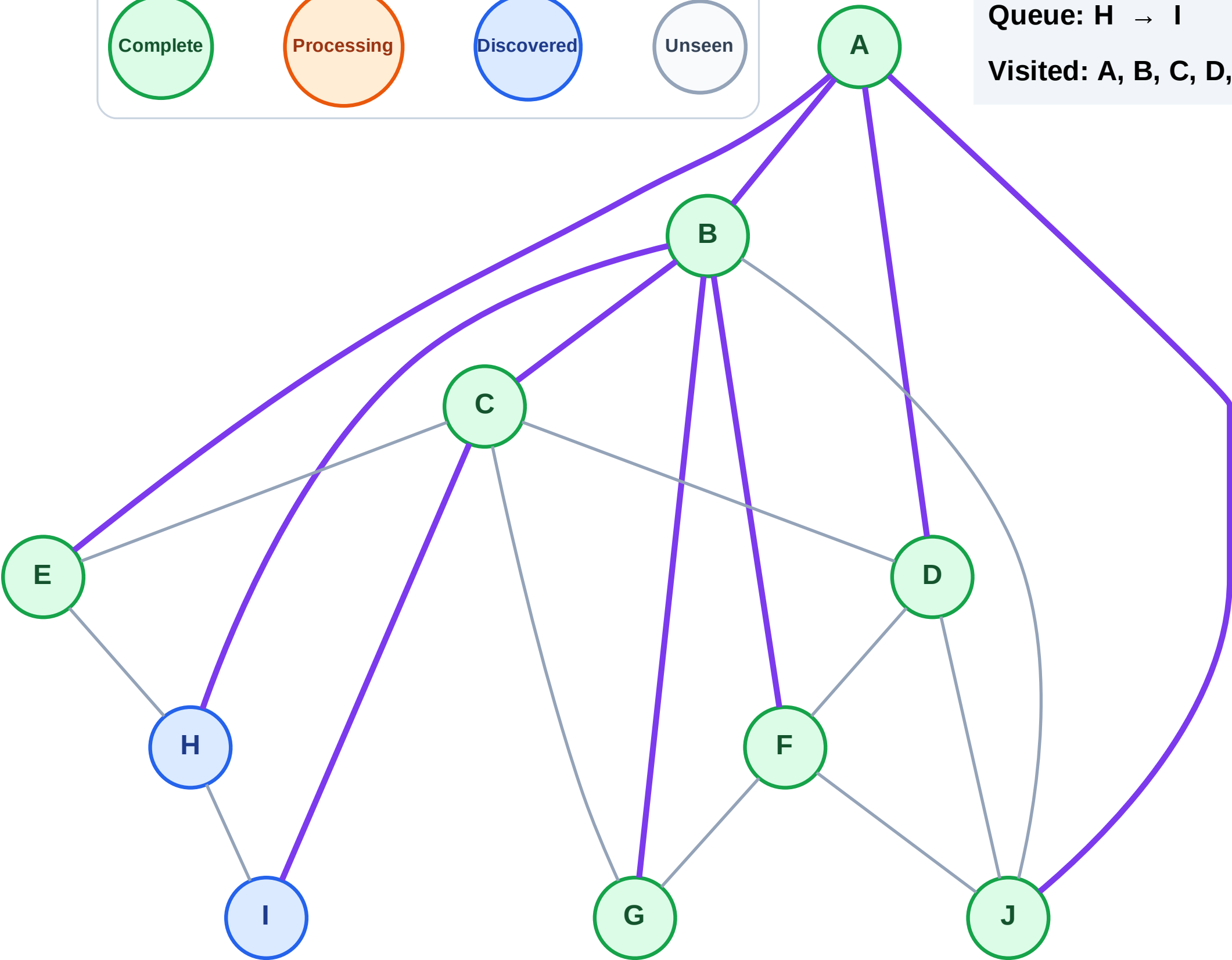
Processing

Discovered

Unseen

Queue: H → I

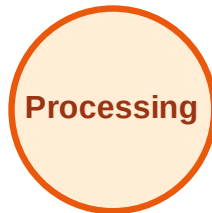
Visited: A, B, C, D, E, F, G, J



BFS Traversal

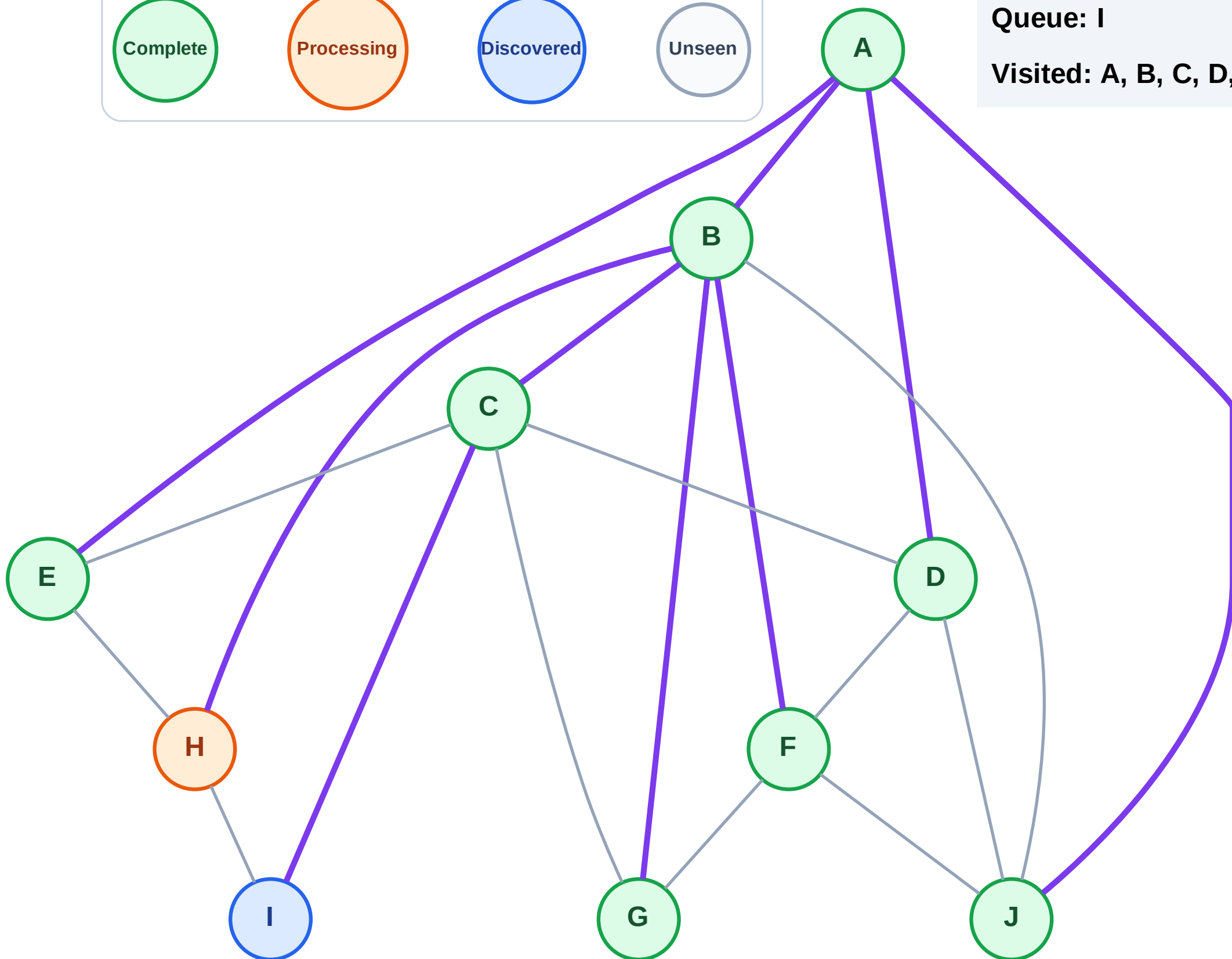
Step 18: Process node H

Legend



Queue: I

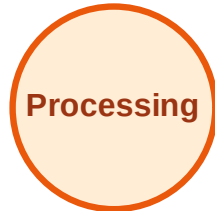
Visited: A, B, C, D, E, F, G, J



BFS Traversal

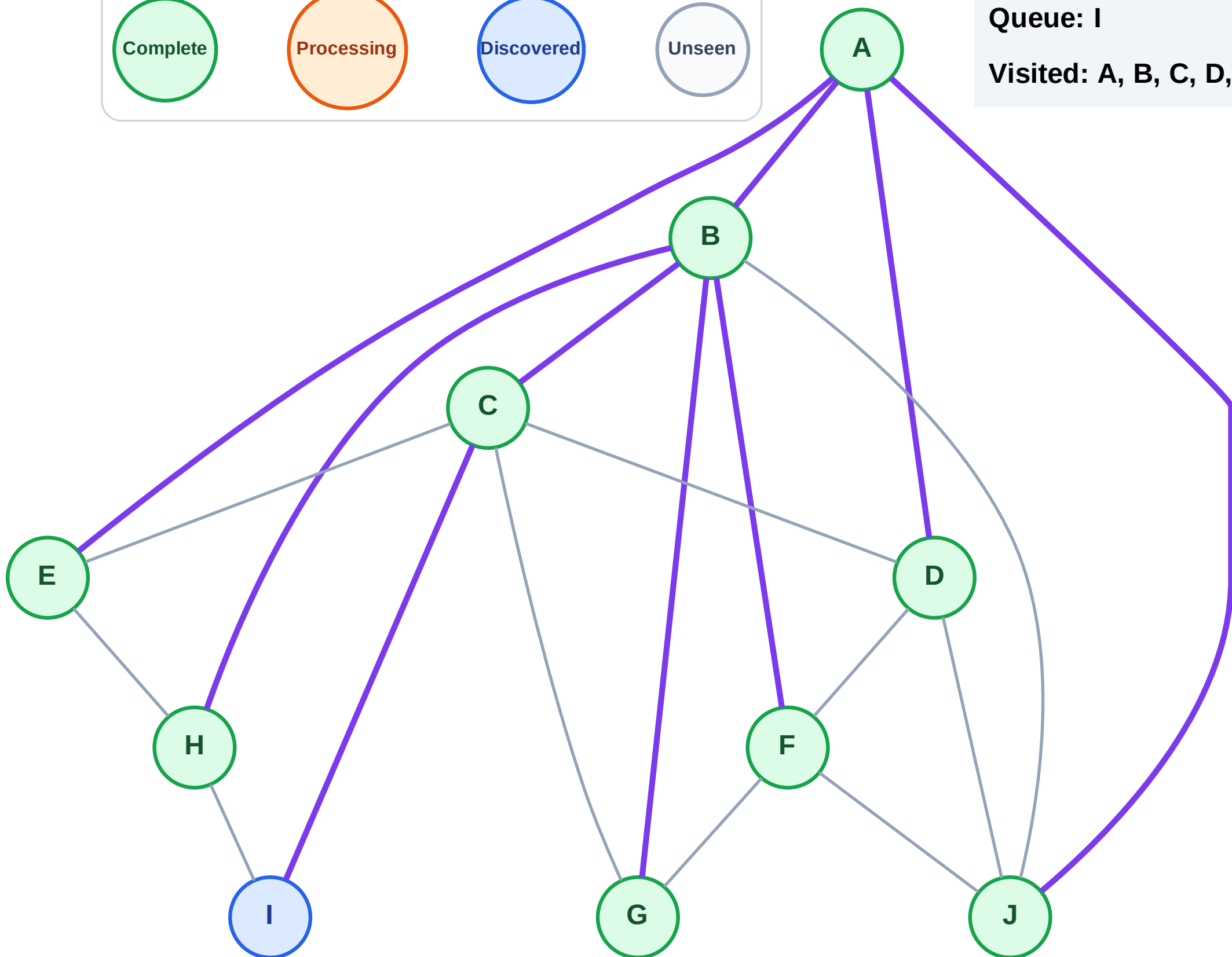
Step 19: No new neighbors from H

Legend



Queue: I

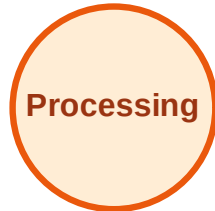
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

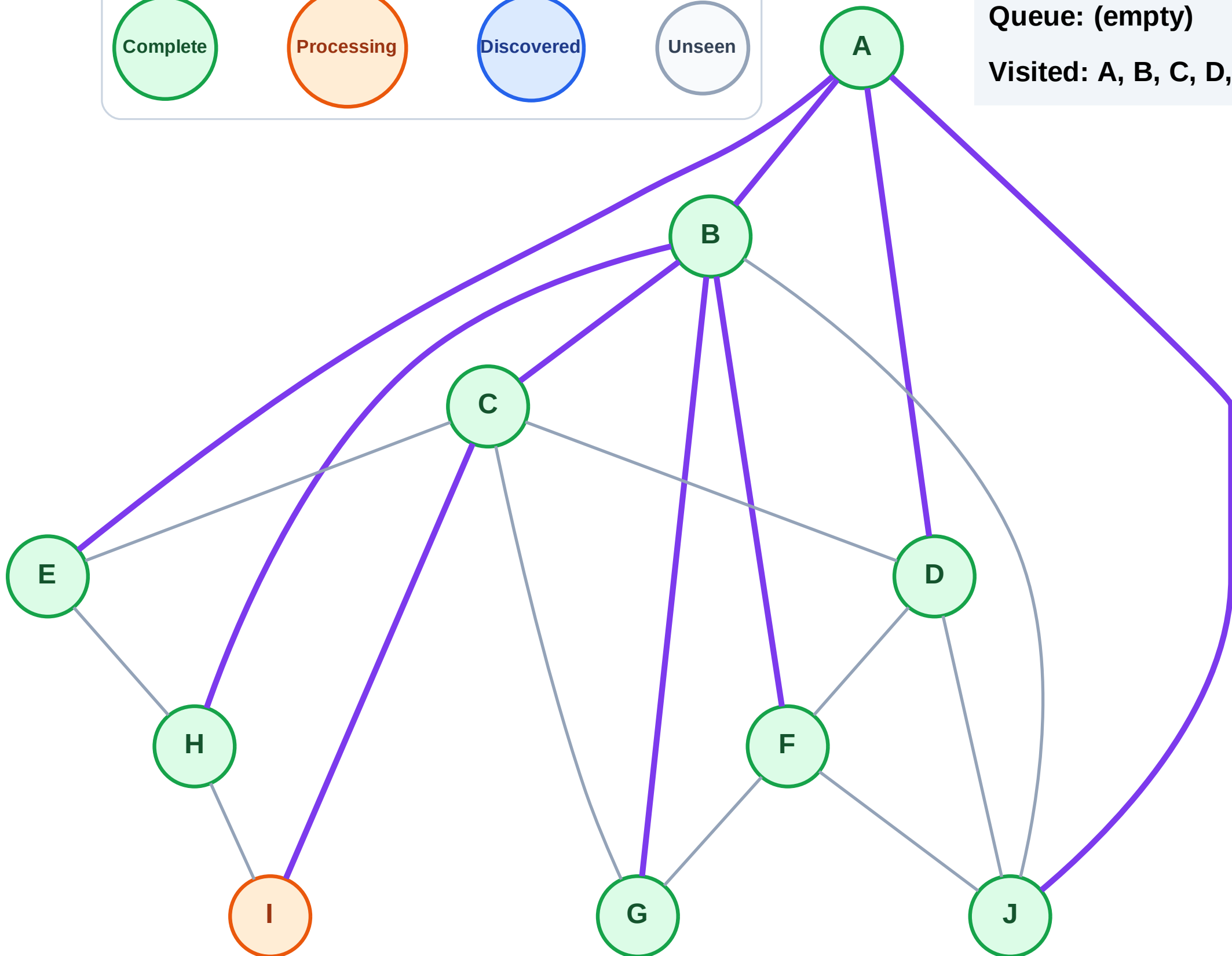
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



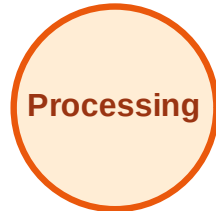
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



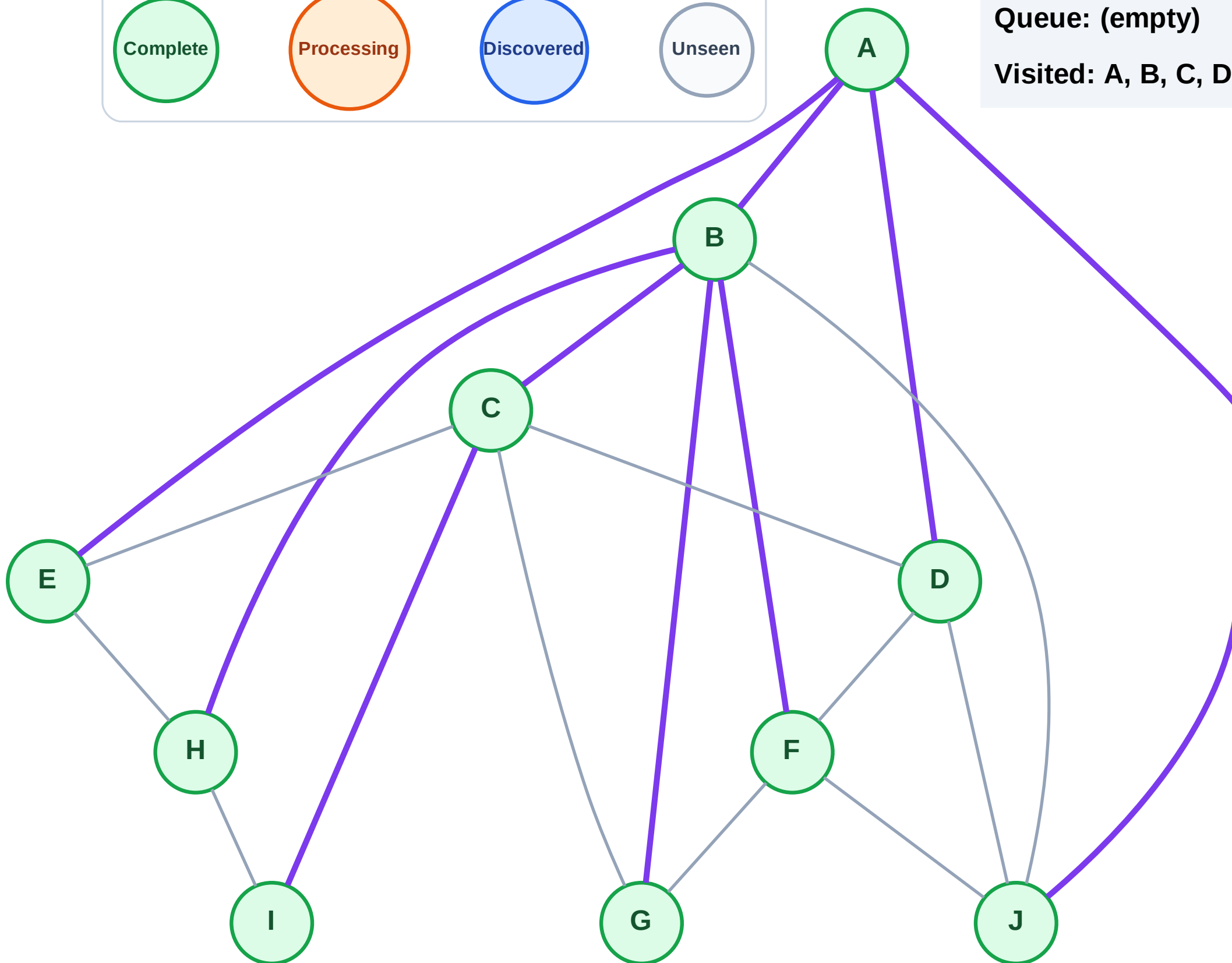
Discovered



Unseen

Queue: (empty)

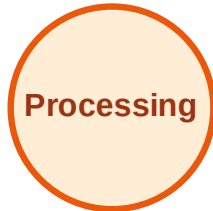
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

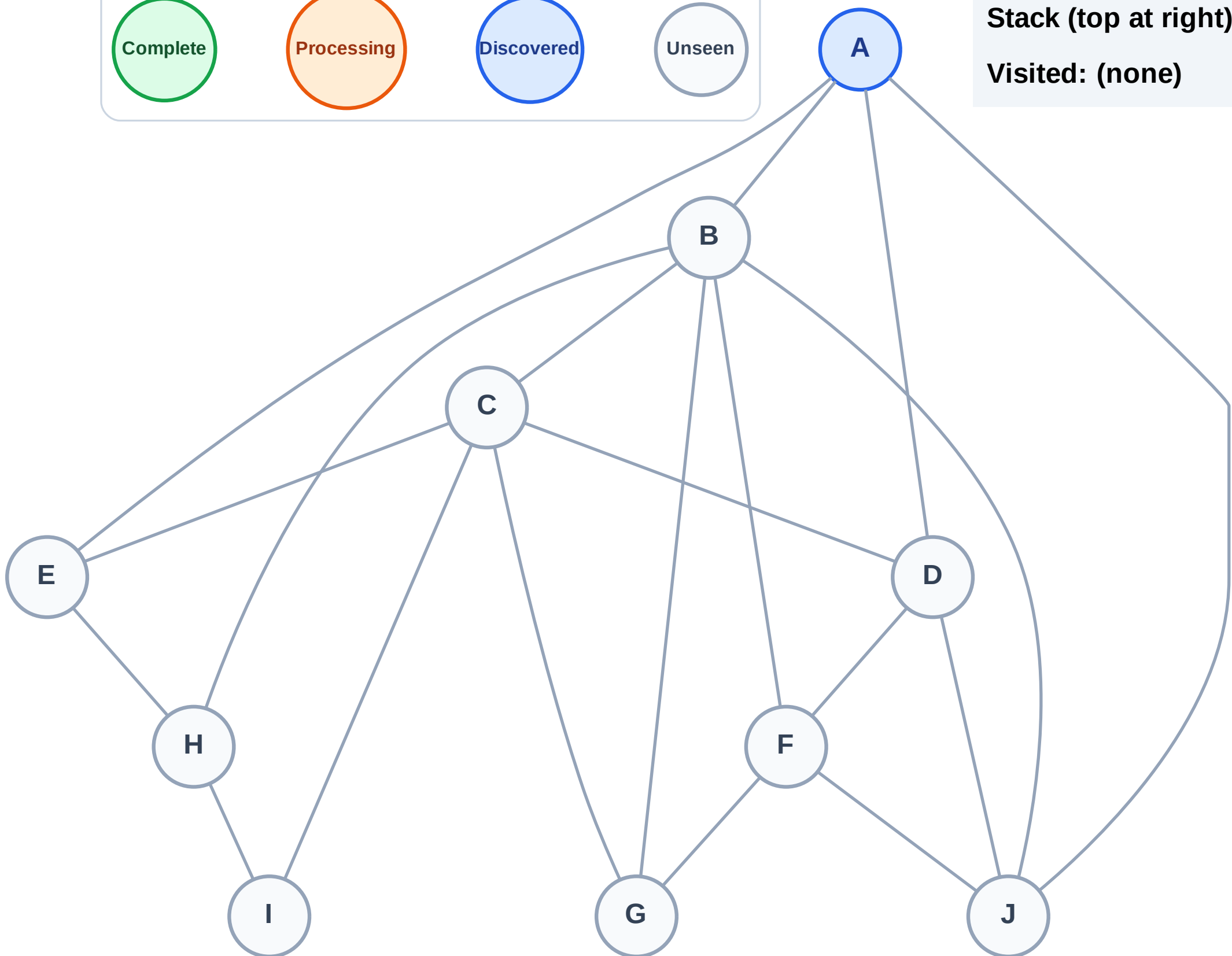
Step 1: Start at A

Legend



Stack (top at right): A

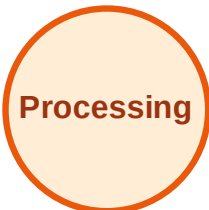
Visited: (none)



DFS Traversal

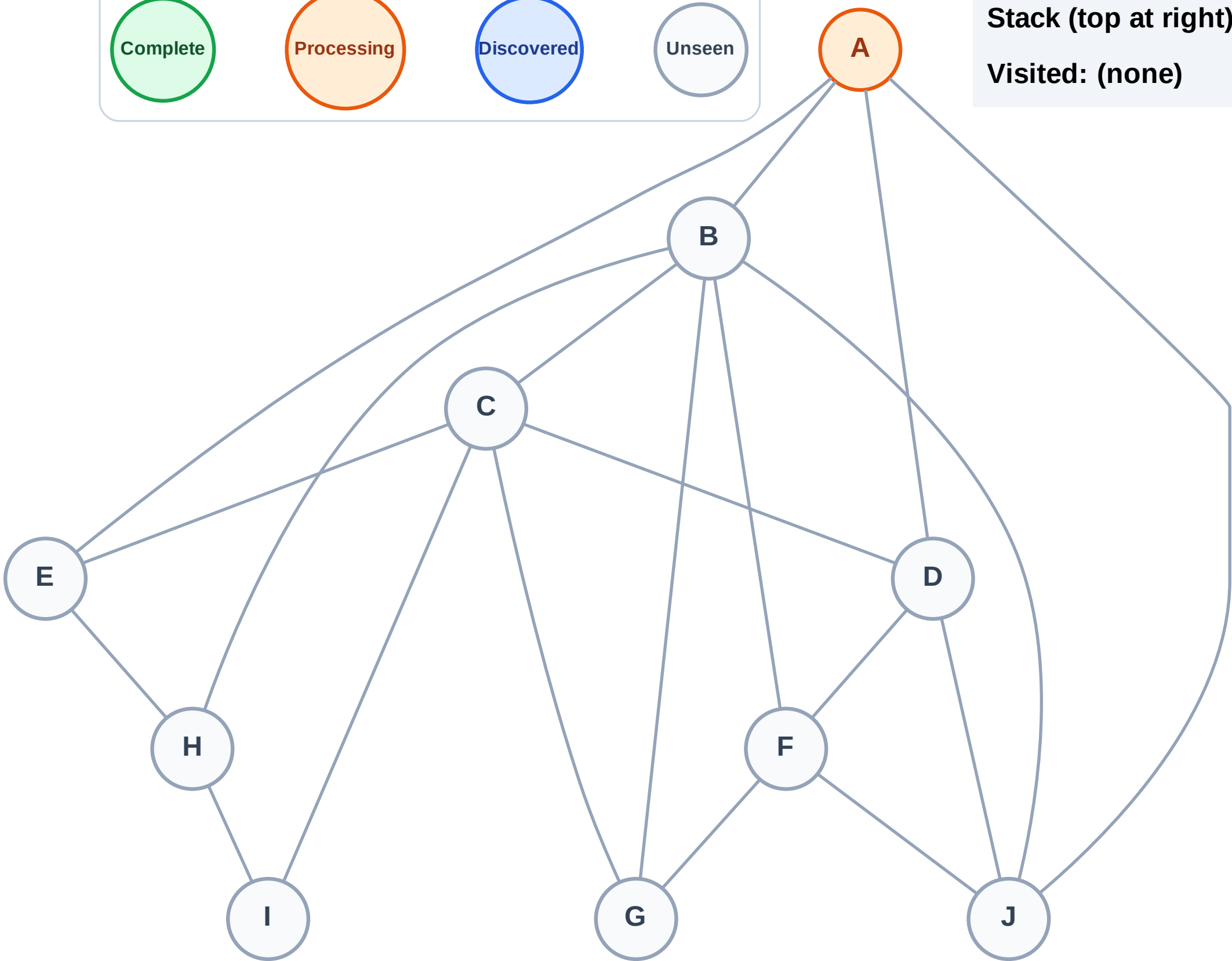
Step 2: Process node A

Legend



Stack (top at right): (empty)

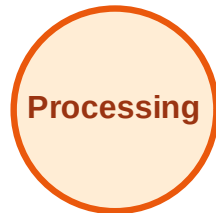
Visited: (none)



DFS Traversal

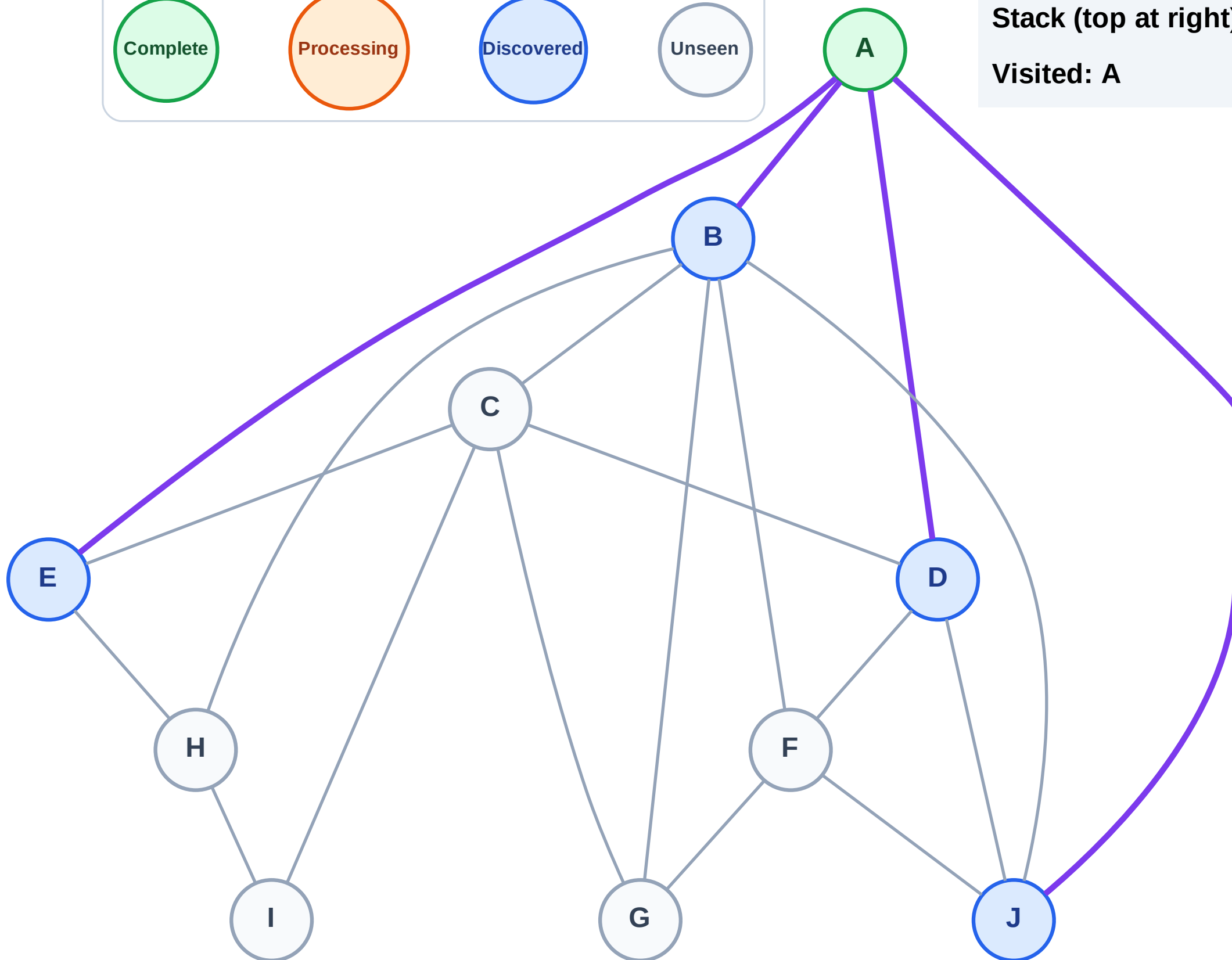
Step 3: Discover J, E, D, B from A

Legend



Stack (top at right): J | E | D | B

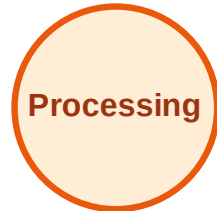
Visited: A



DFS Traversal

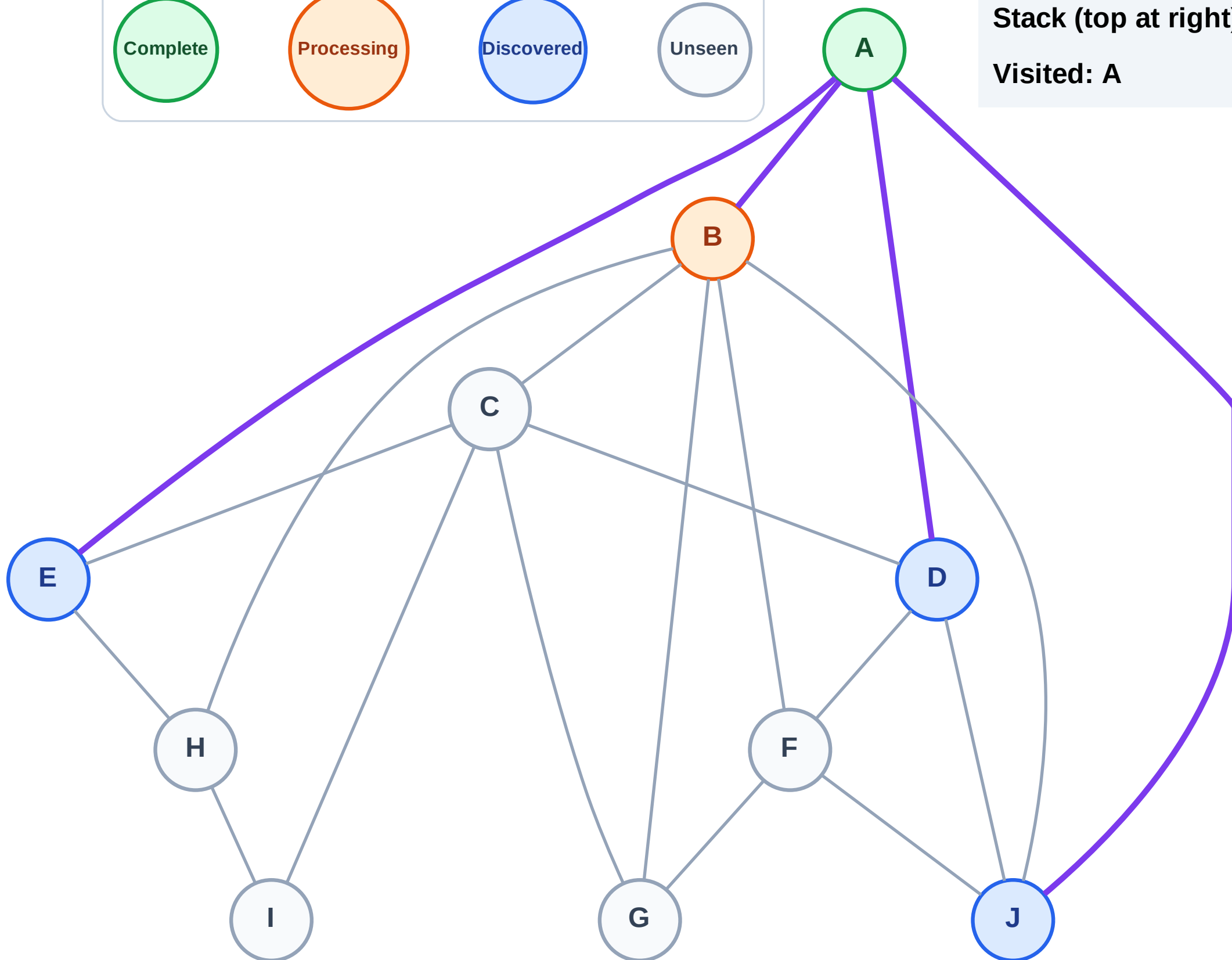
Step 4: Process node B

Legend



Stack (top at right): J | E | D

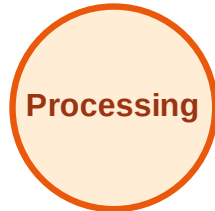
Visited: A



DFS Traversal

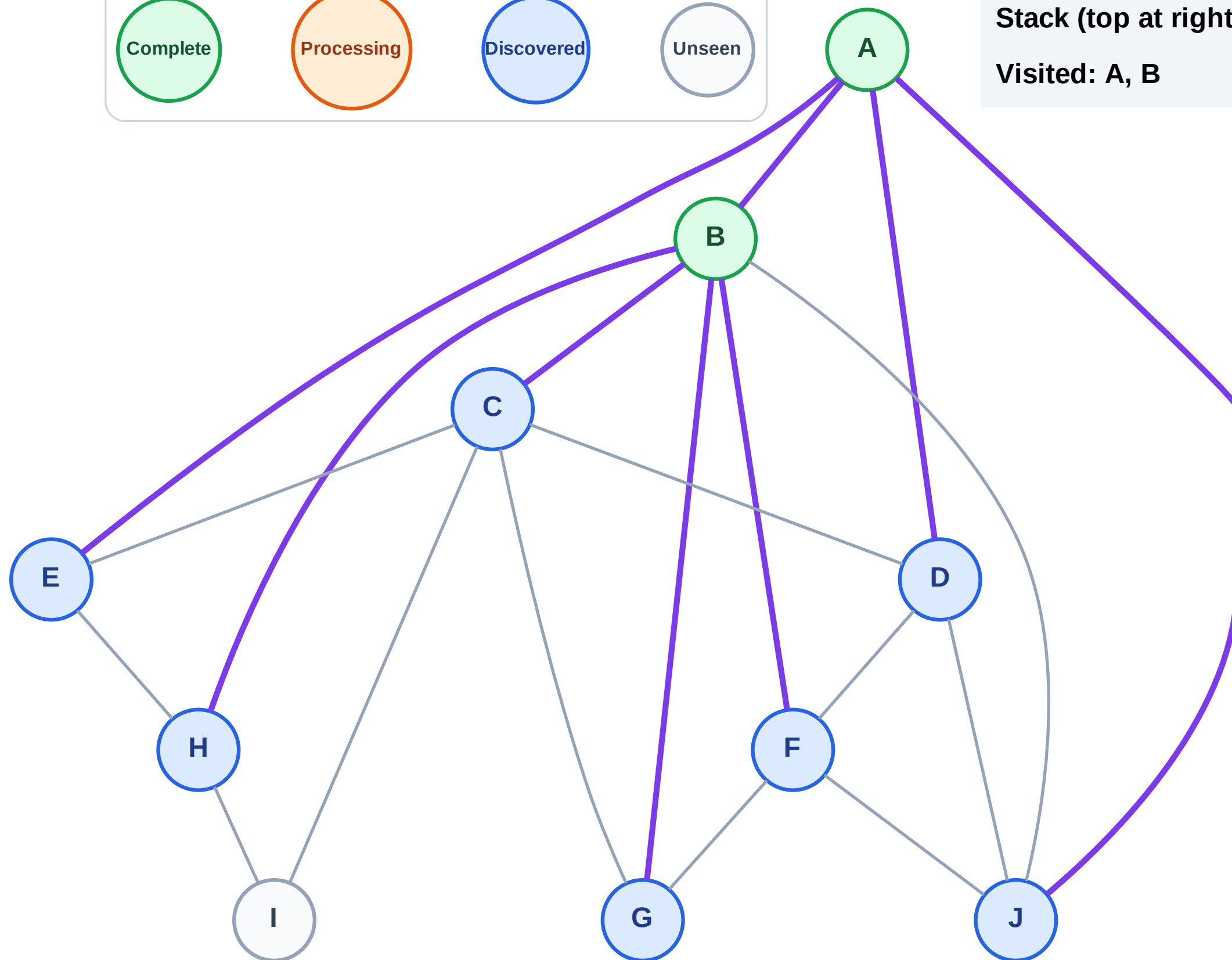
Step 5: Discover H, G, F, C from B

Legend

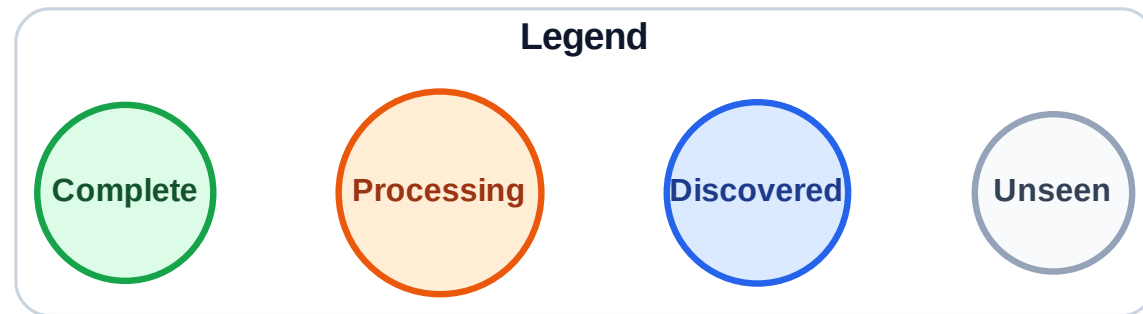


Stack (top at right): J | E | D | H | G | F | C

Visited: A, B

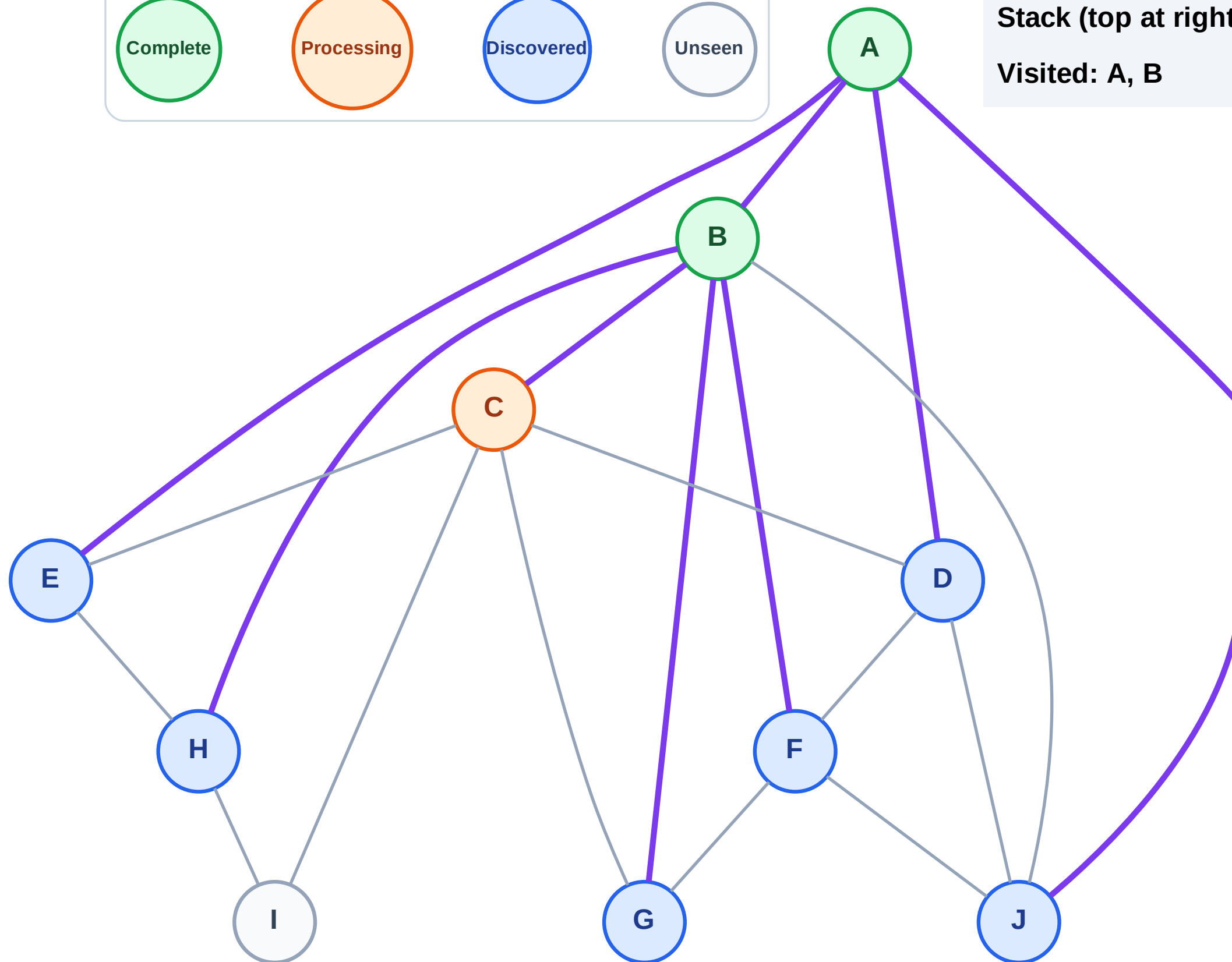


Step 6: Process node C



Stack (top at right): J | E | D | H | G | F

Visited: A, B



DFS Traversal

Step 7: Discover I from C

Legend

Complete

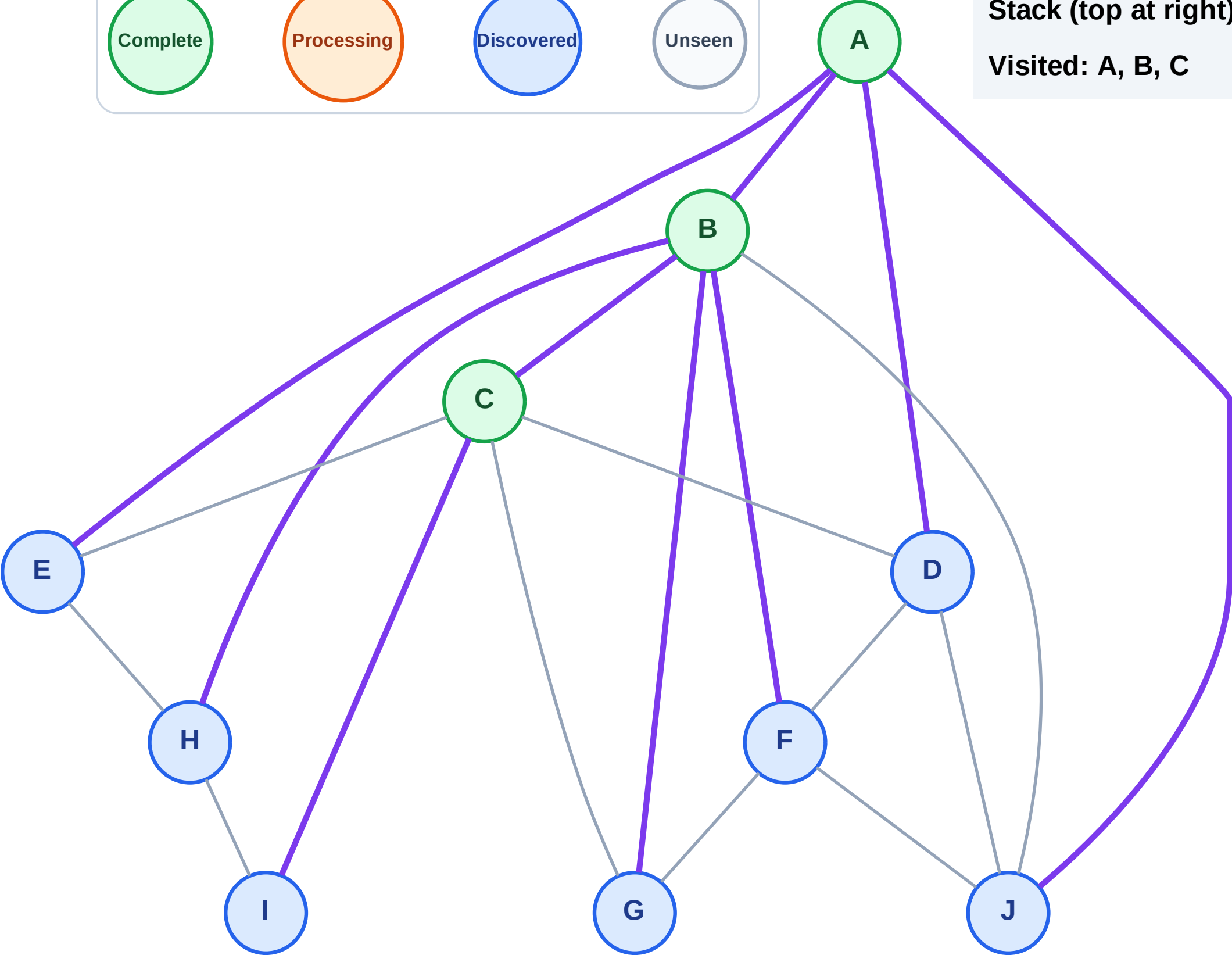
Processing

Discovered

Unseen

Stack (top at right): J | E | D | H | G | F | I

Visited: A, B, C



DFS Traversal

Step 8: Process node I

Legend

Complete

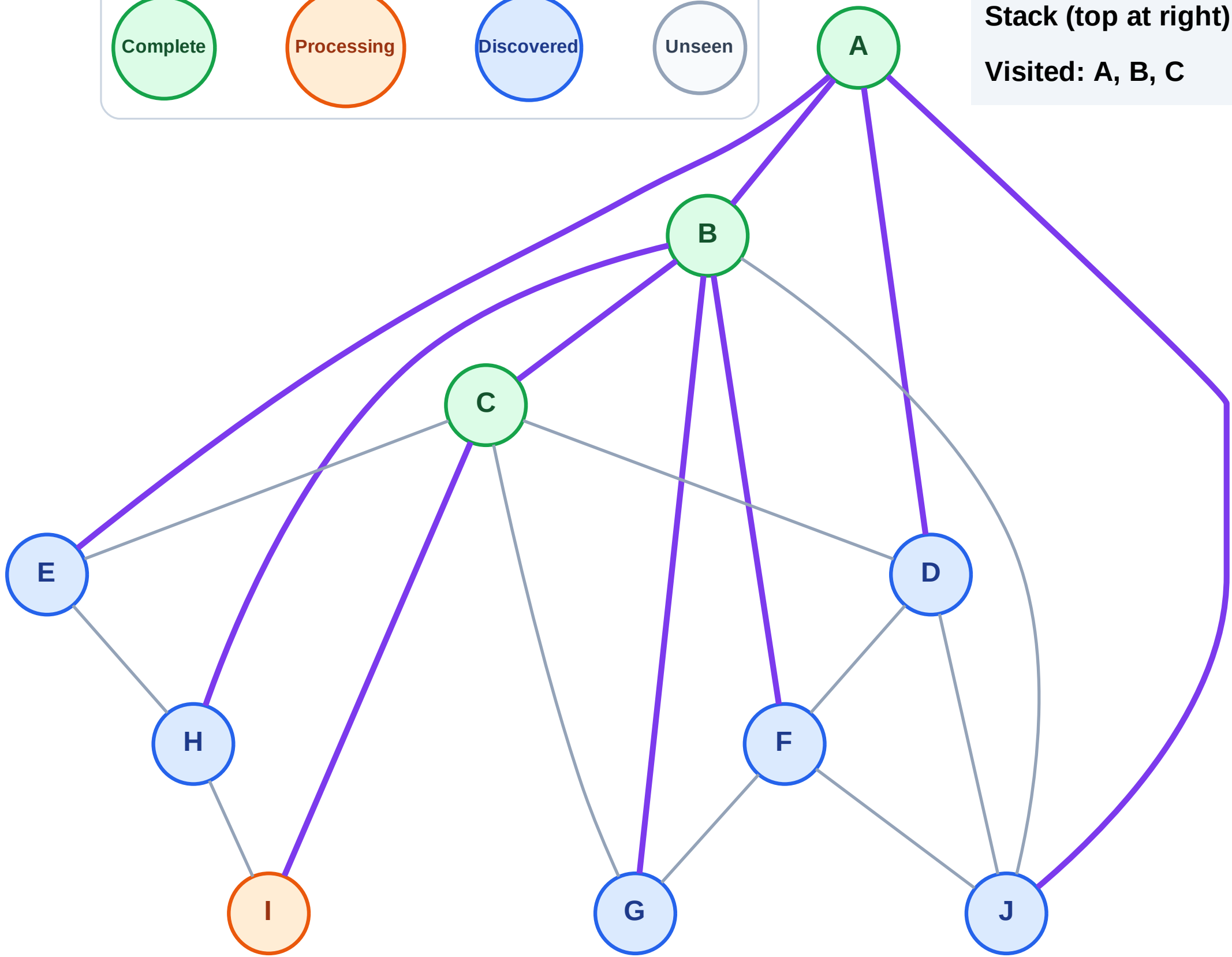
Processing

Discovered

Unseen

Stack (top at right): J | E | D | H | G | F

Visited: A, B, C

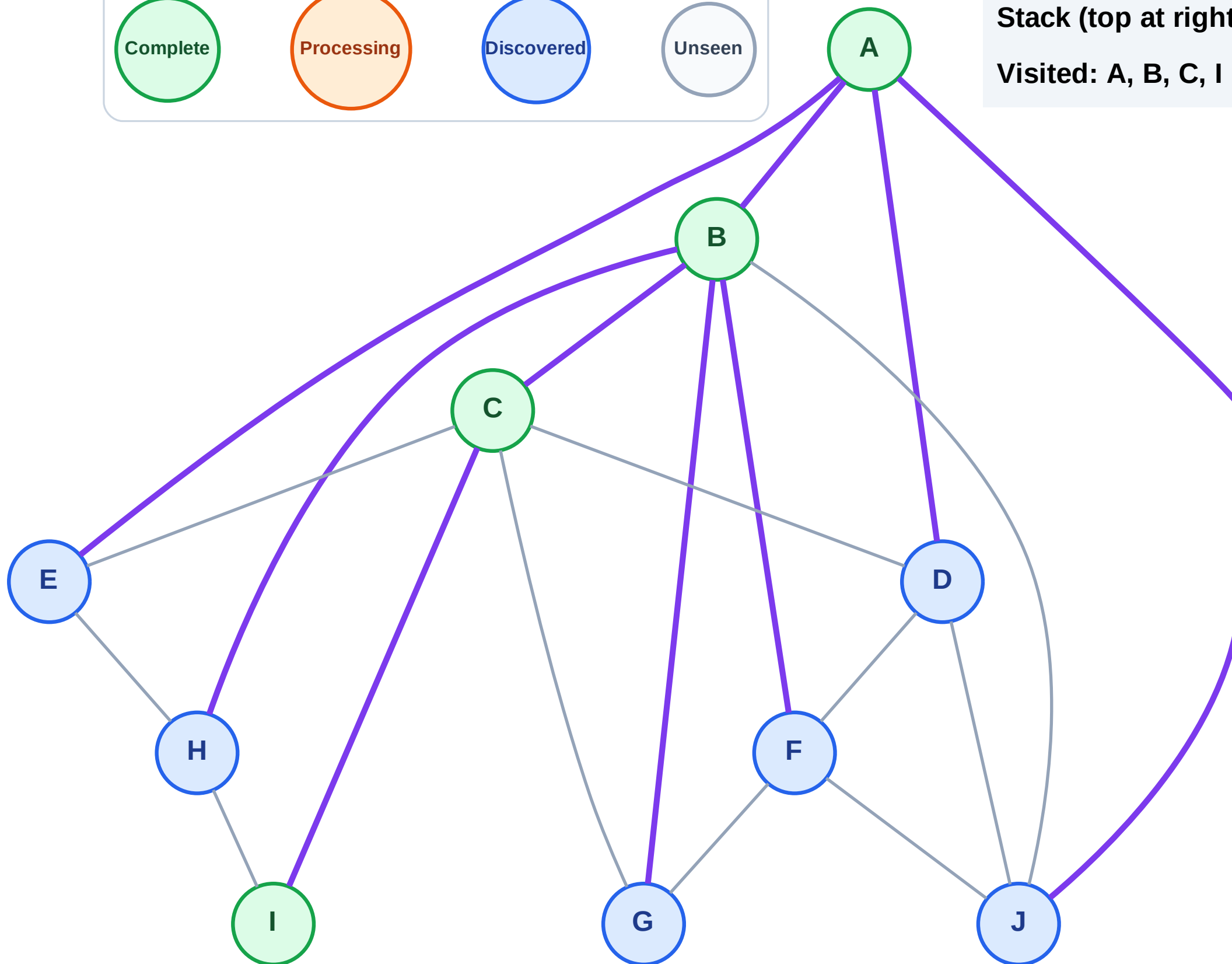


Step 9: No new neighbors from I

Legend

Unseen

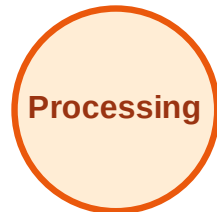
Visited: A, B, C, I



DFS Traversal

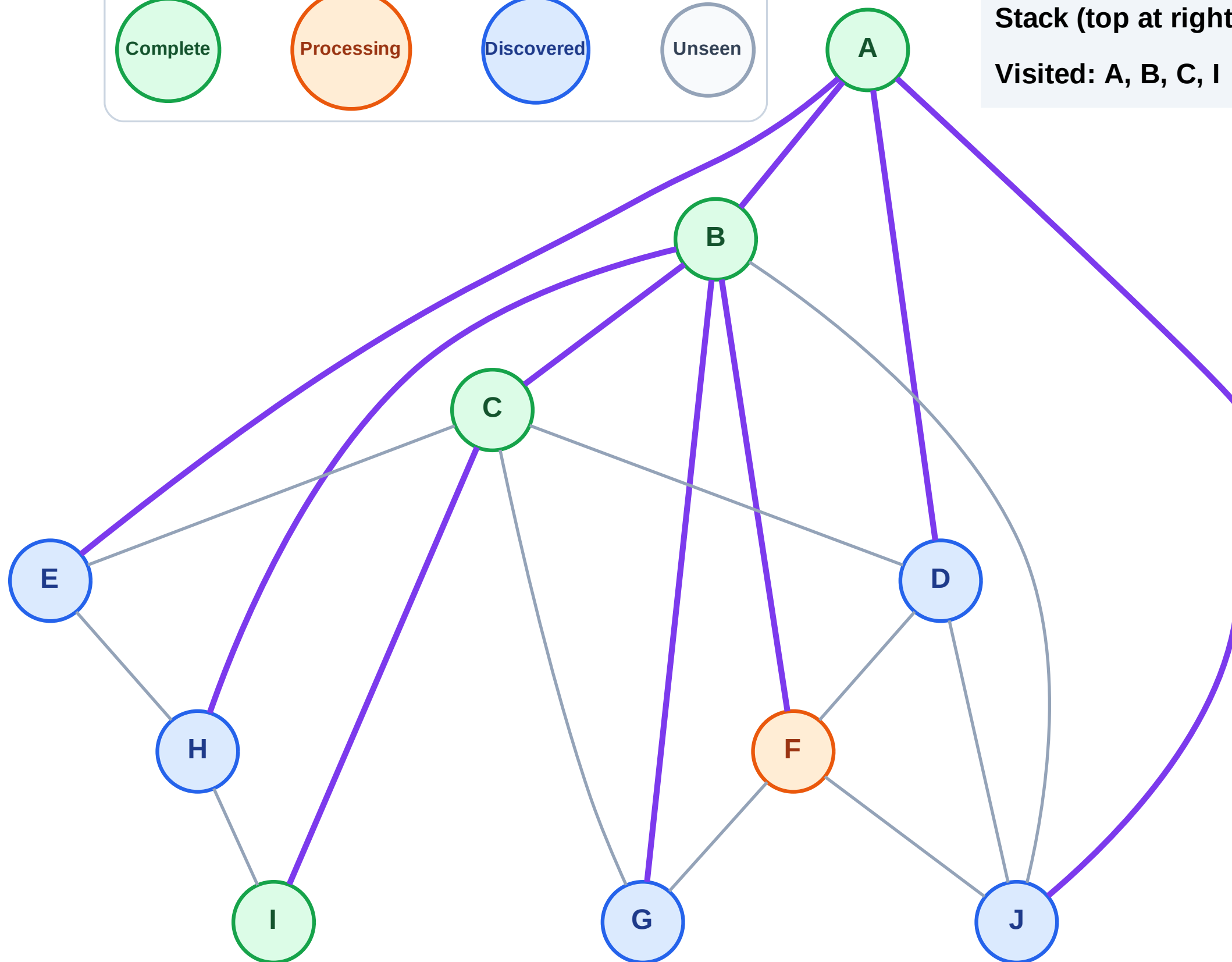
Step 10: Process node F

Legend



Stack (top at right): J | E | D | H | G

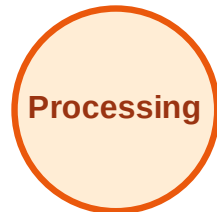
Visited: A, B, C, I



DFS Traversal

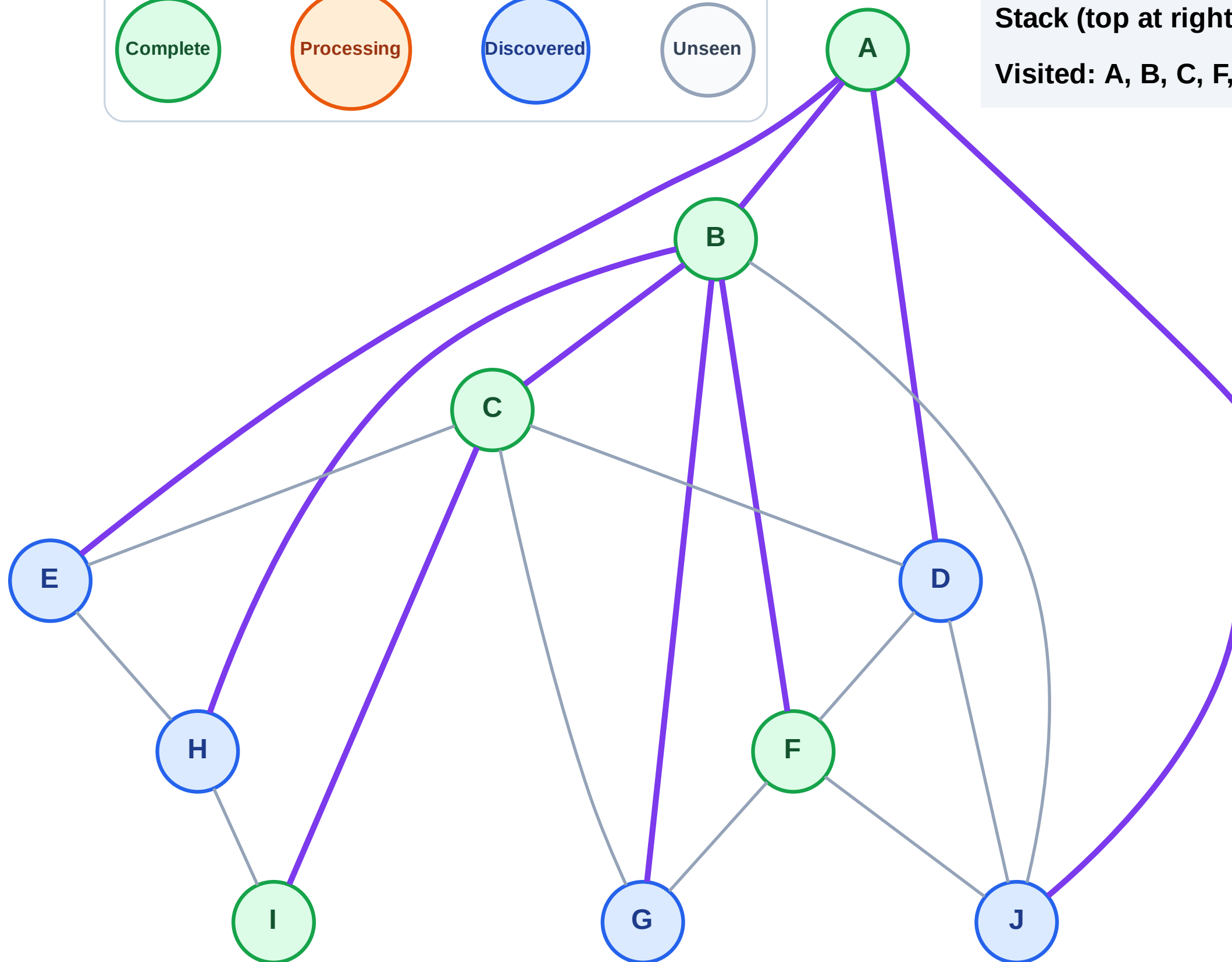
Step 11: No new neighbors from F

Legend



Stack (top at right): J | E | D | H | G

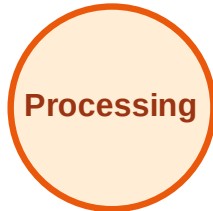
Visited: A, B, C, F, I



DFS Traversal

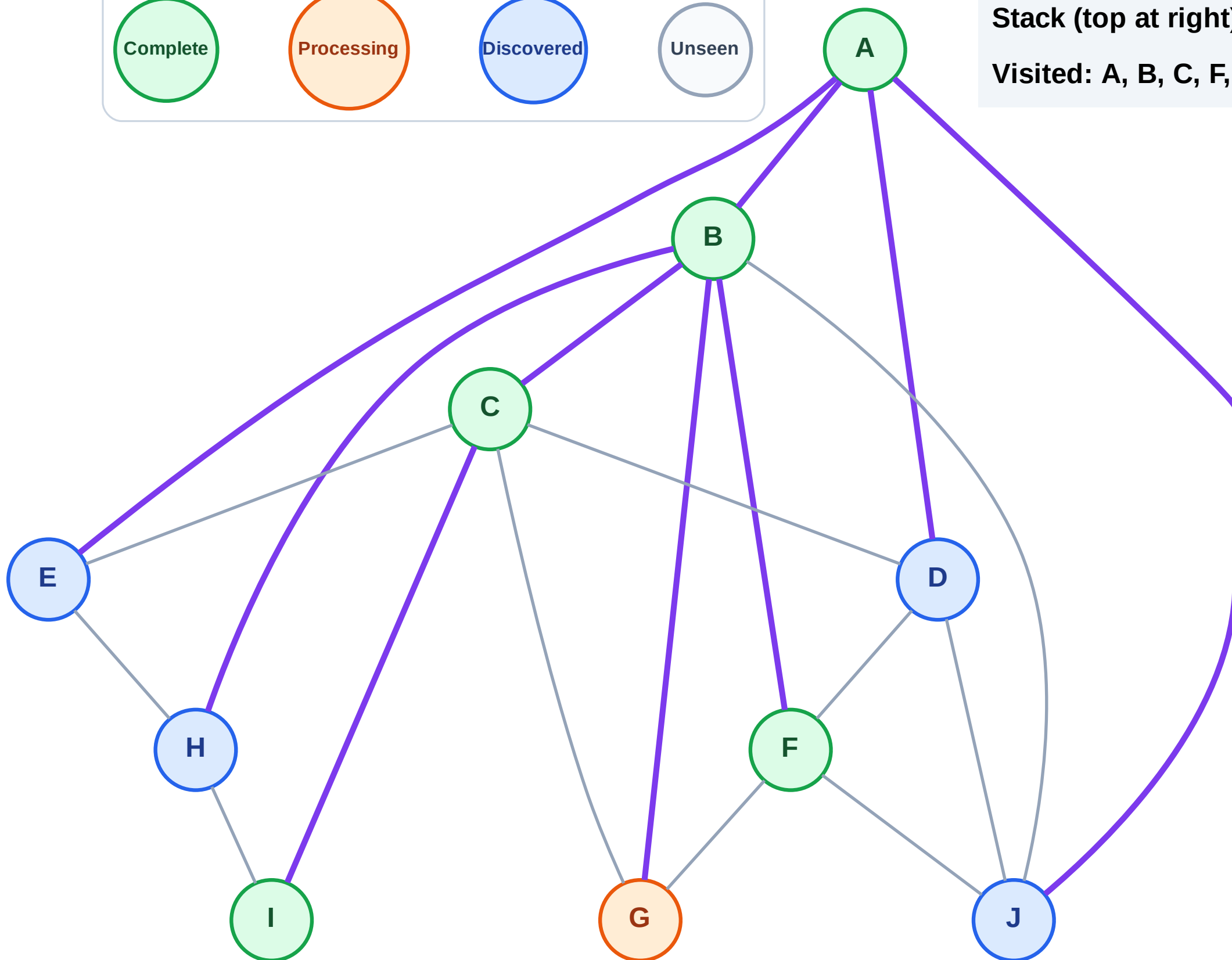
Step 12: Process node G

Legend



Stack (top at right): J | E | D | H

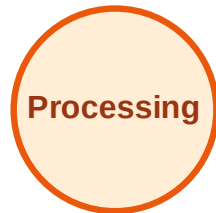
Visited: A, B, C, F, I



DFS Traversal

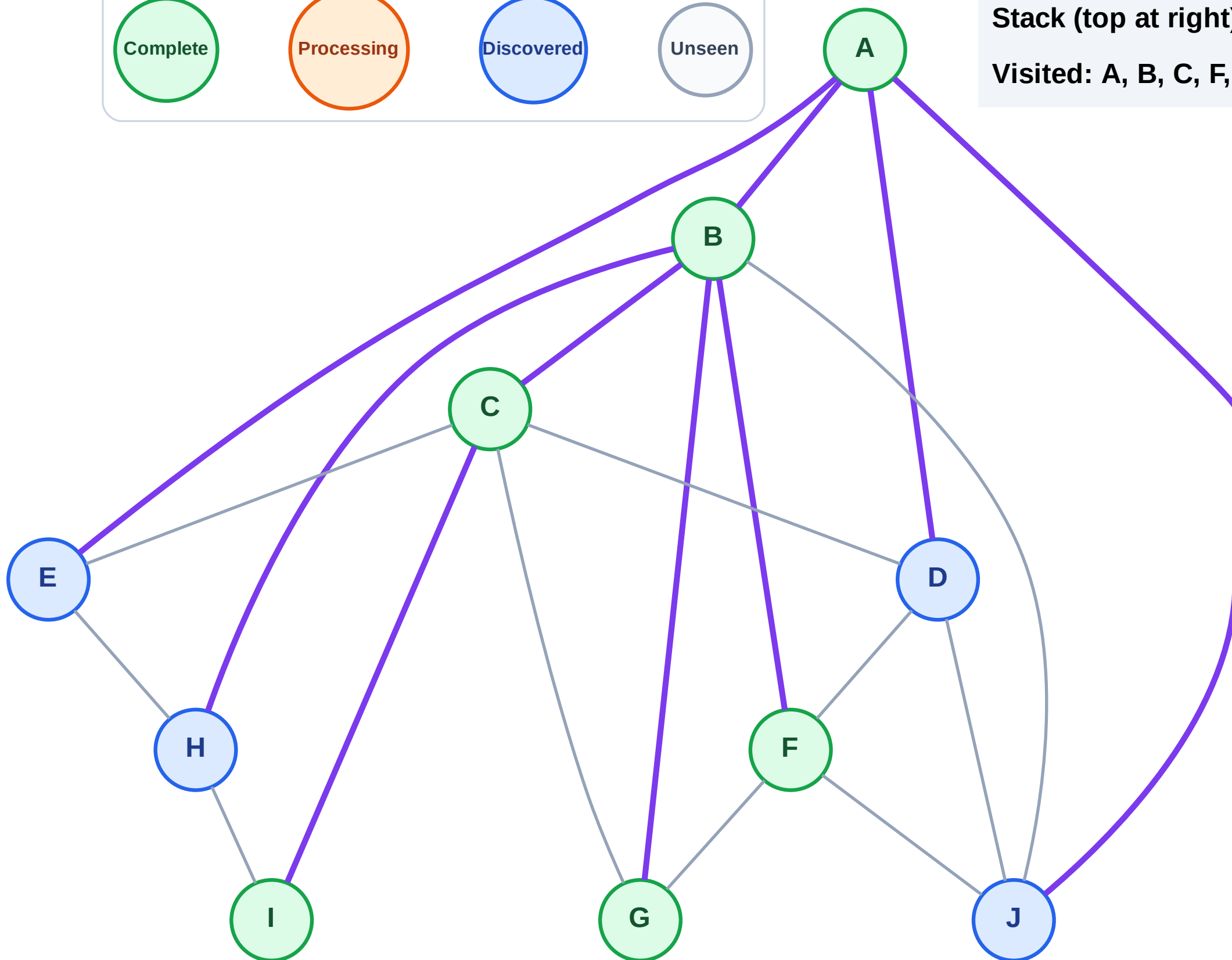
Step 13: No new neighbors from G

Legend



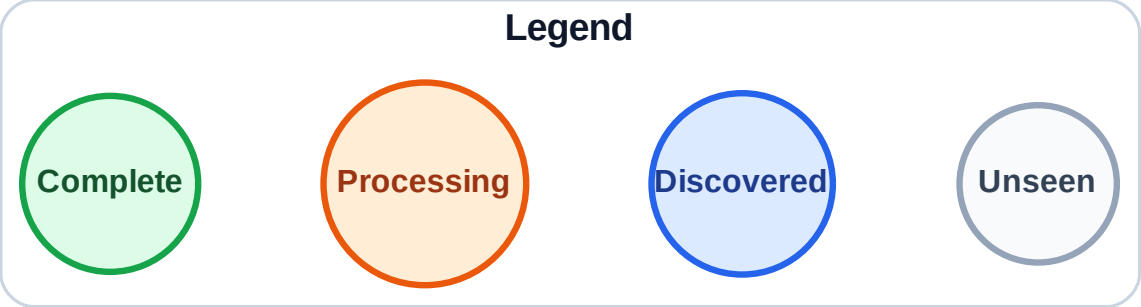
Stack (top at right): J | E | D | H

Visited: A, B, C, F, G, I

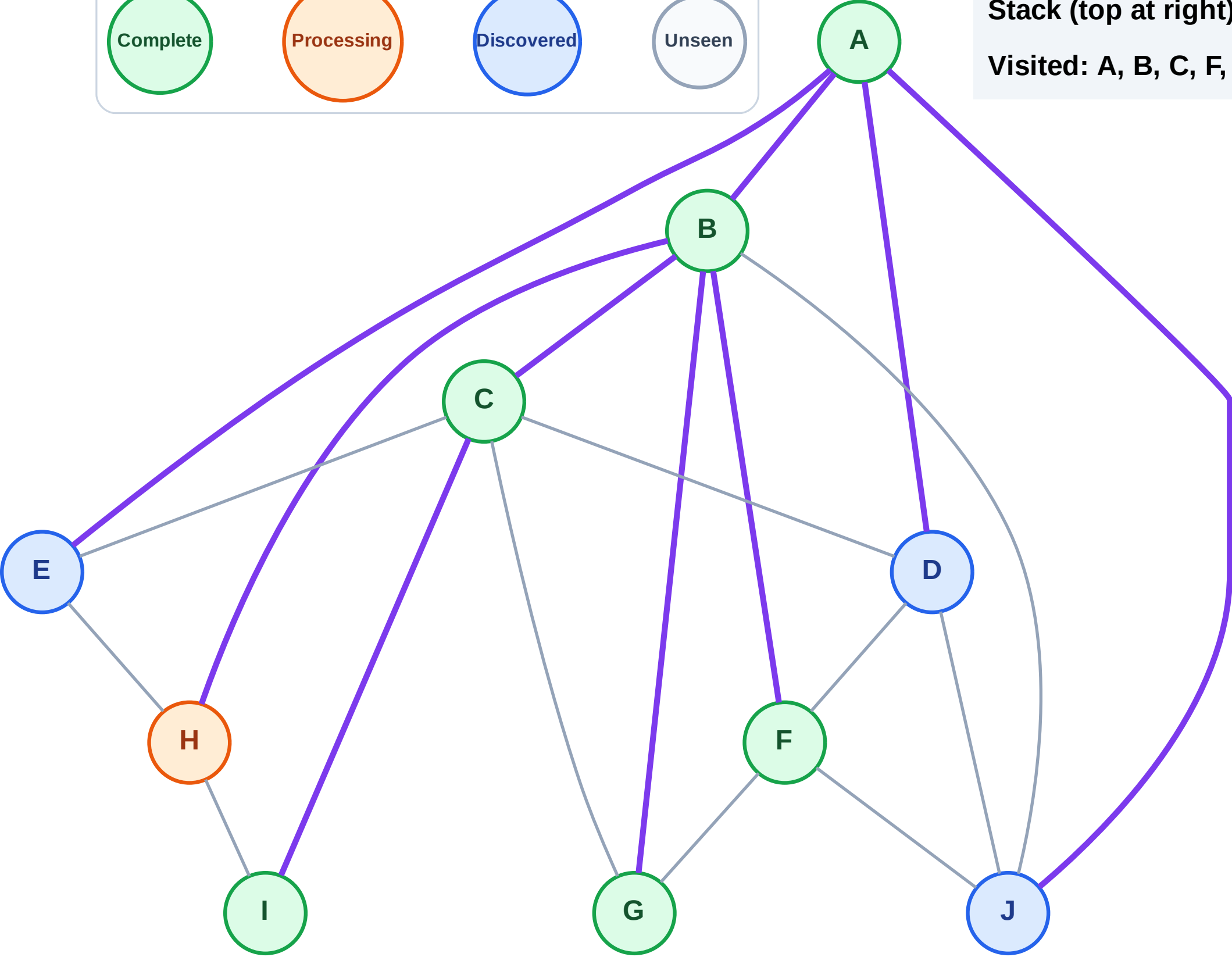


DFS Traversal

Step 14: Process node H



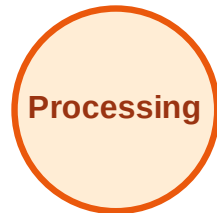
Stack (top at right): J | E | D
Visited: A, B, C, F, G, I



DFS Traversal

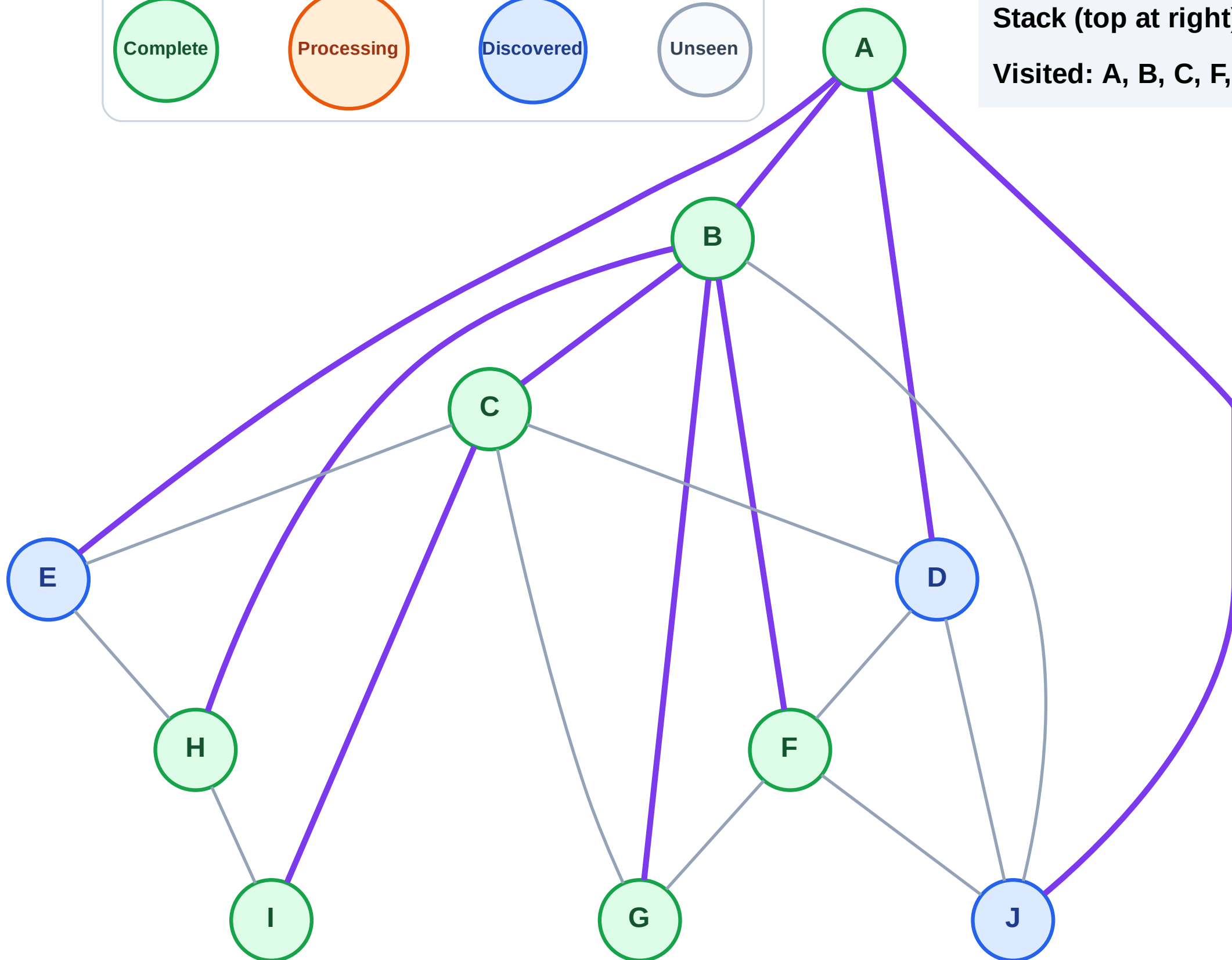
Step 15: No new neighbors from H

Legend



Stack (top at right): J | E | D

Visited: A, B, C, F, G, H, I



DFS Traversal

Step 16: Process node D

Legend

Complete

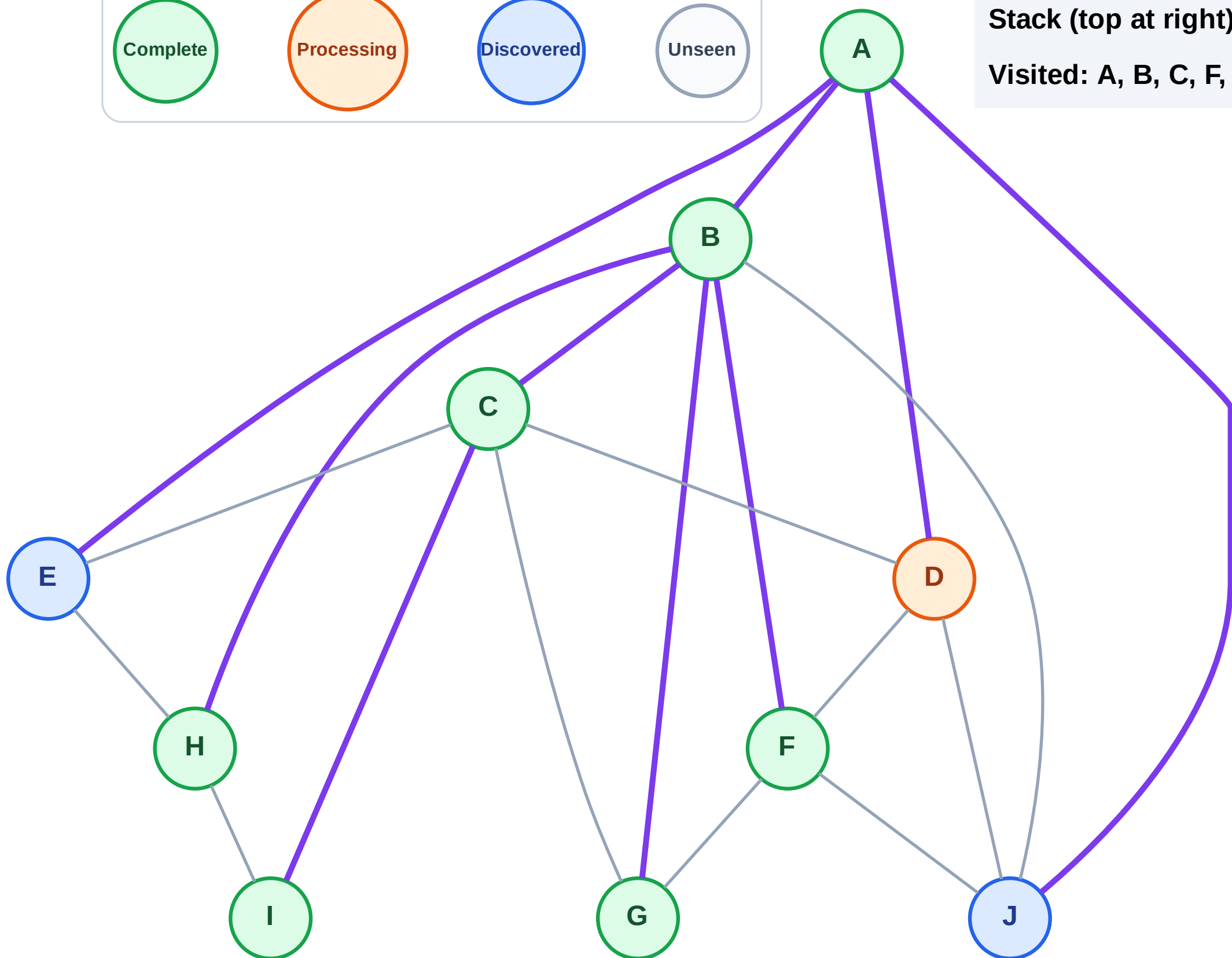
Processing

Discovered

Unseen

Stack (top at right): J | E

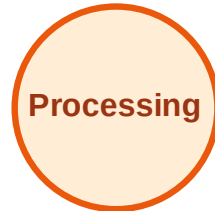
Visited: A, B, C, F, G, H, I



DFS Traversal

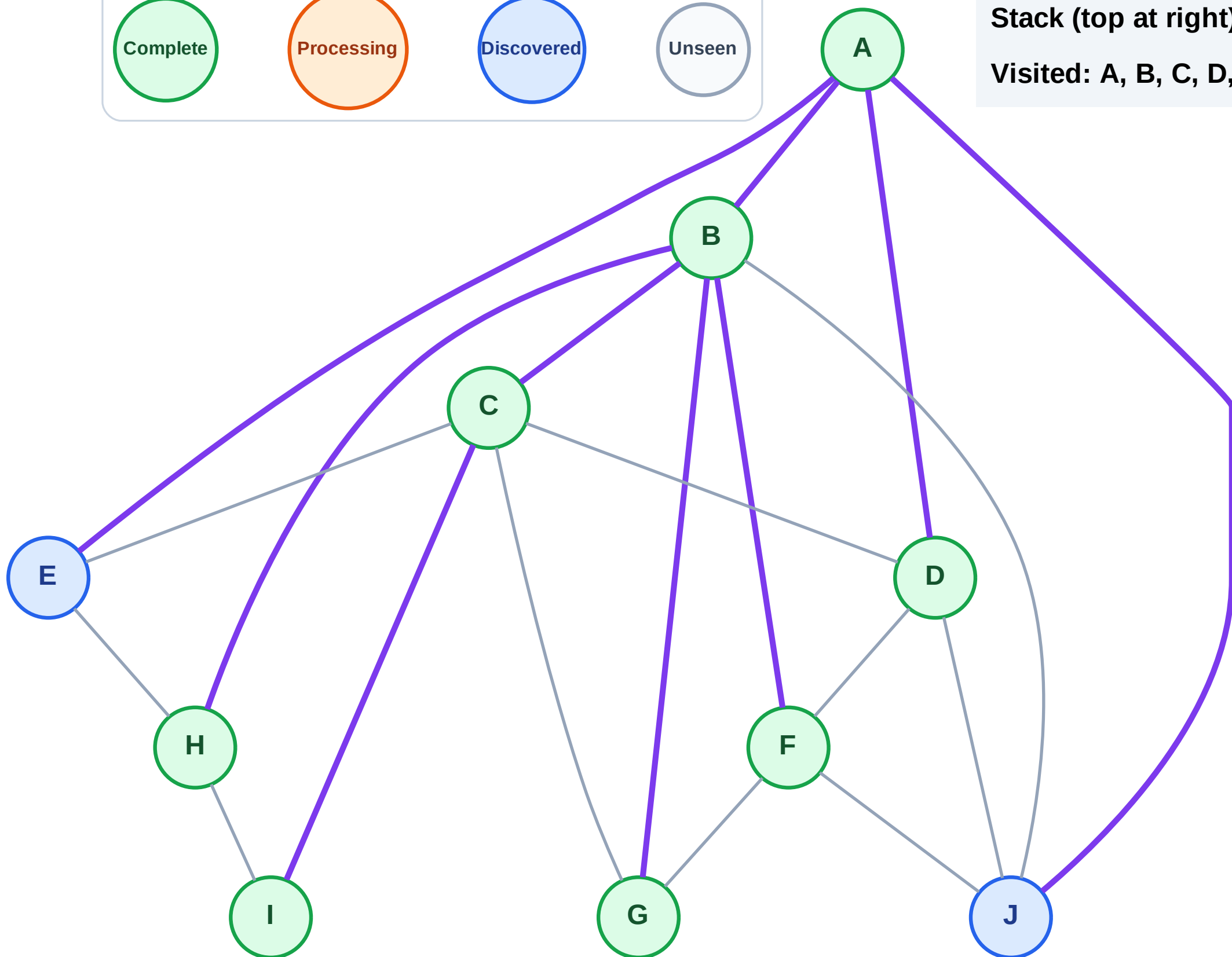
Step 17: No new neighbors from D

Legend



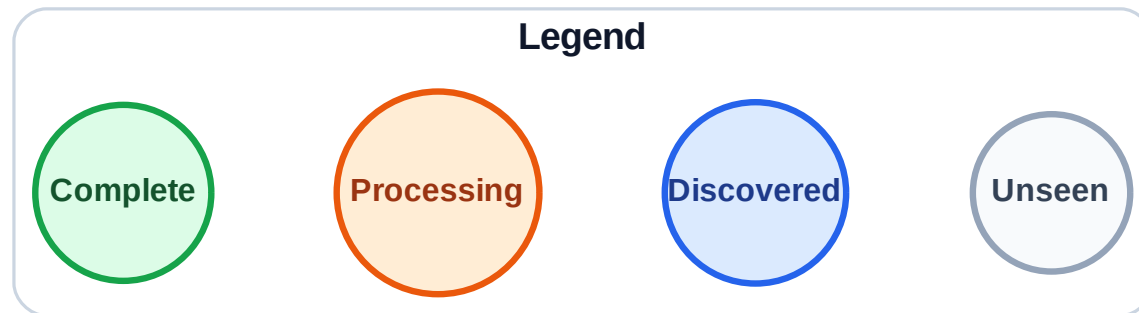
Stack (top at right): J | E

Visited: A, B, C, D, F, G, H, I



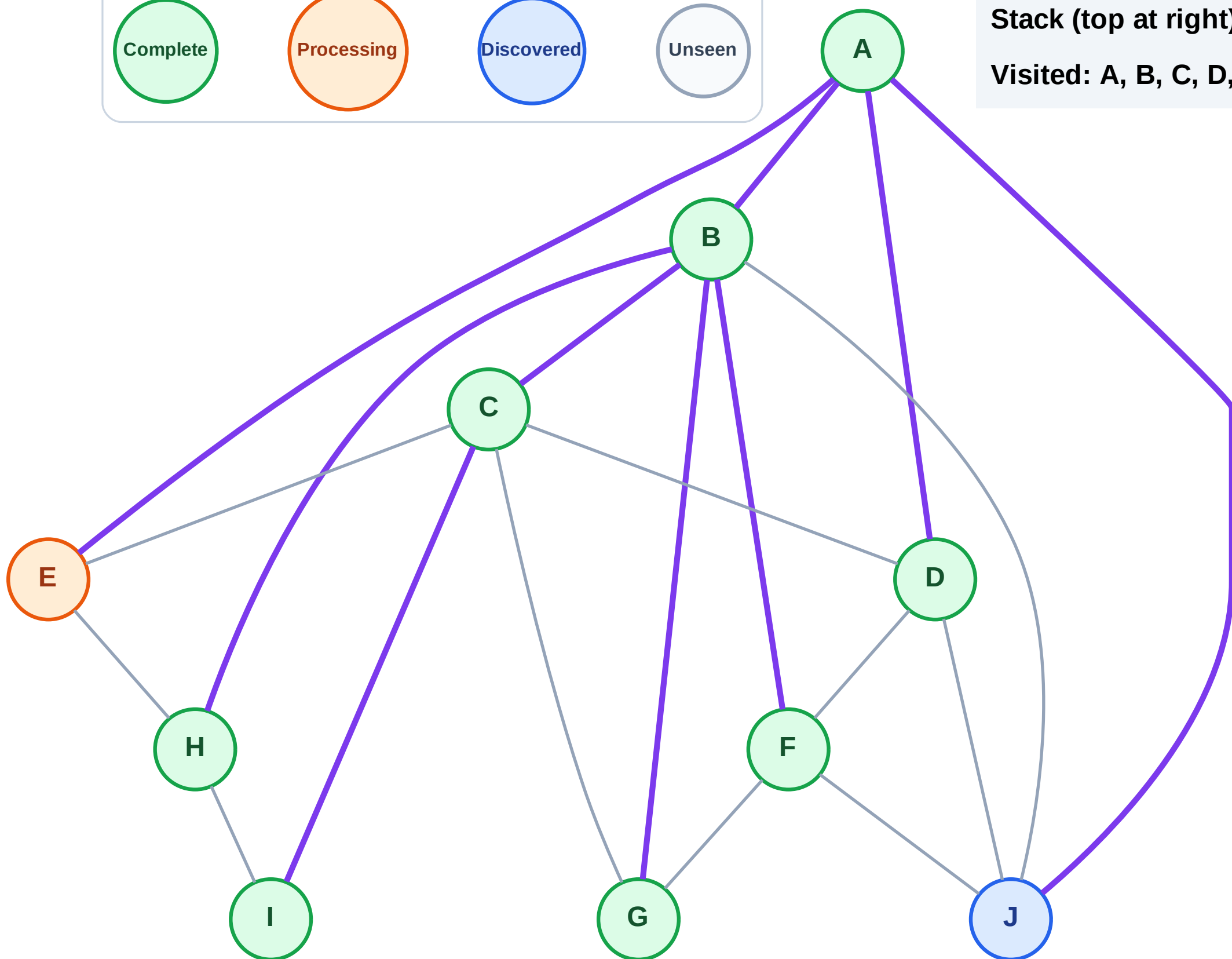
Step 18: Process node E

Step 18: Process node E



Stack (top at right): J

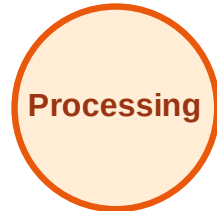
Visited: A, B, C, D, F, G, H, I



DFS Traversal

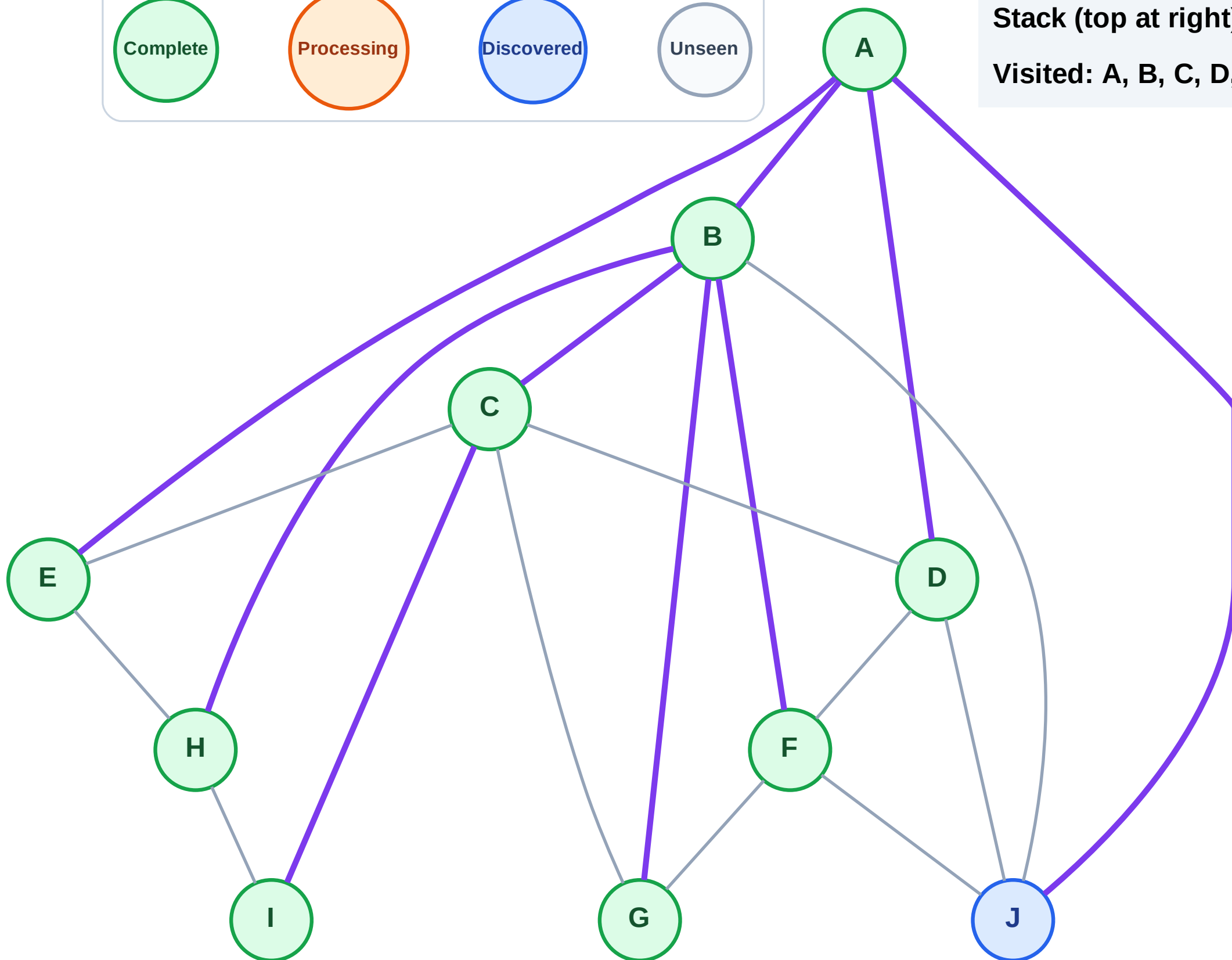
Step 19: No new neighbors from E

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

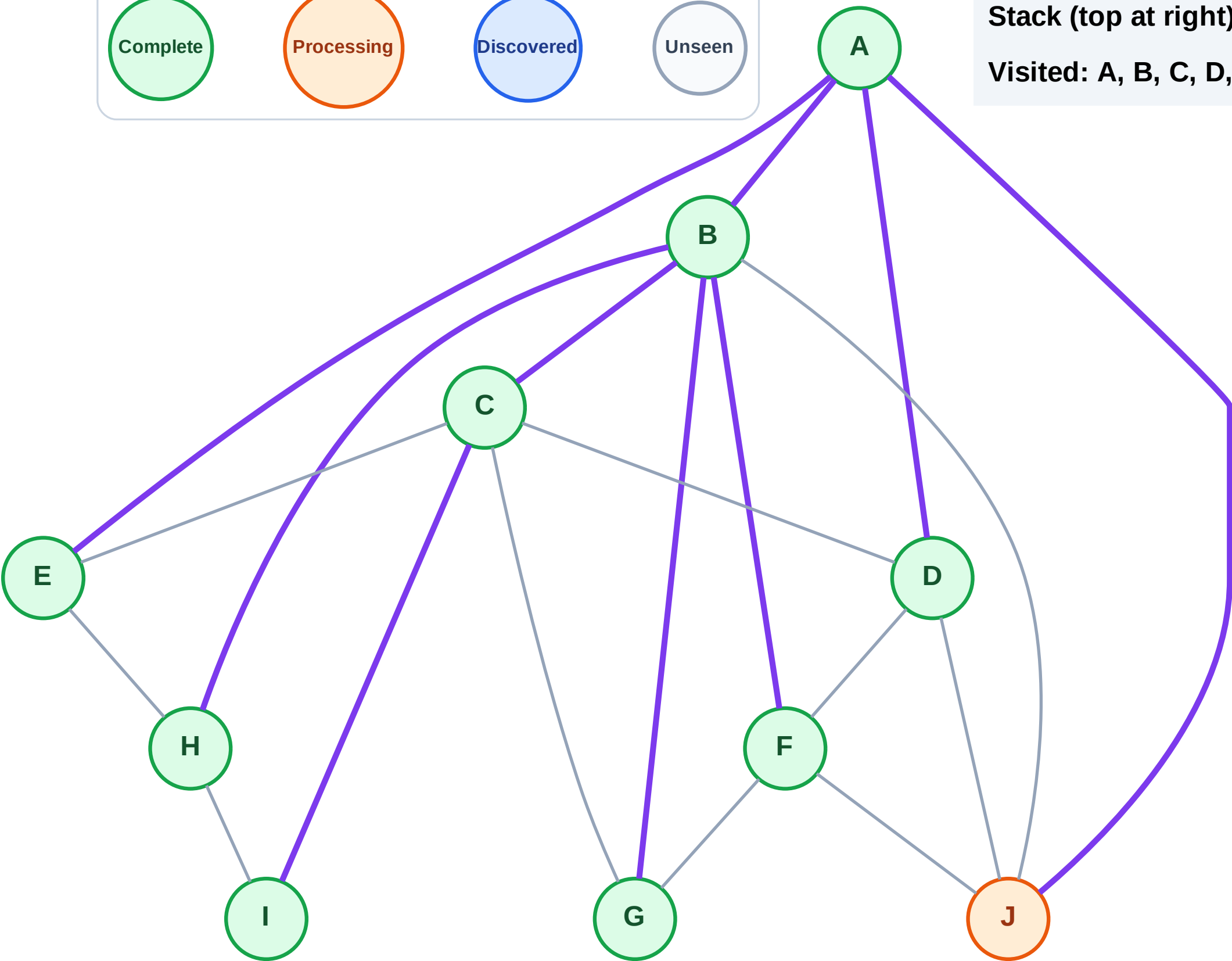
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



Step 21: No new neighbors from J

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, I, J

